



Master On Sustainable Cities, 2025

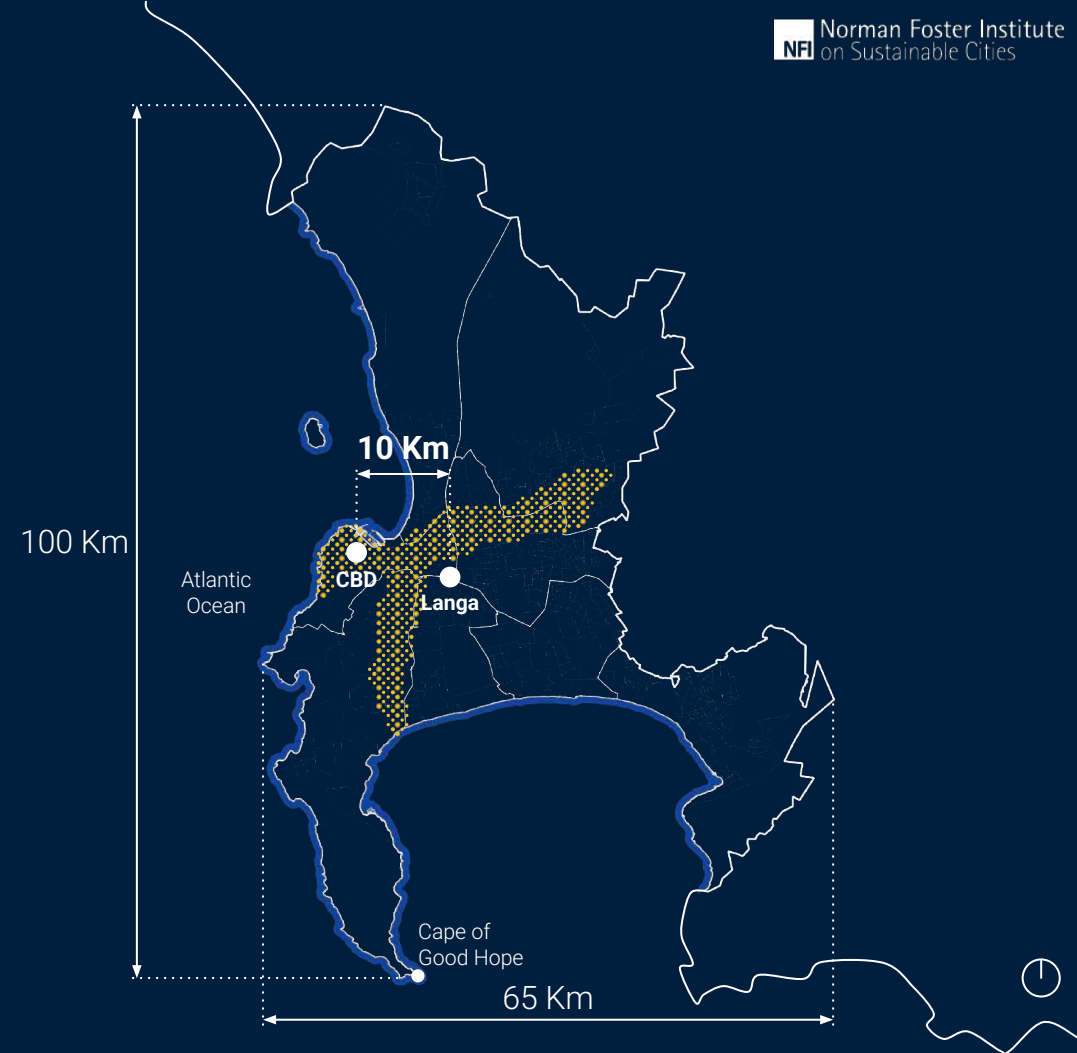


Central Business District
Cape Town



Township of Langa
Cape Town

A Geography of Opportunity



Employment Density

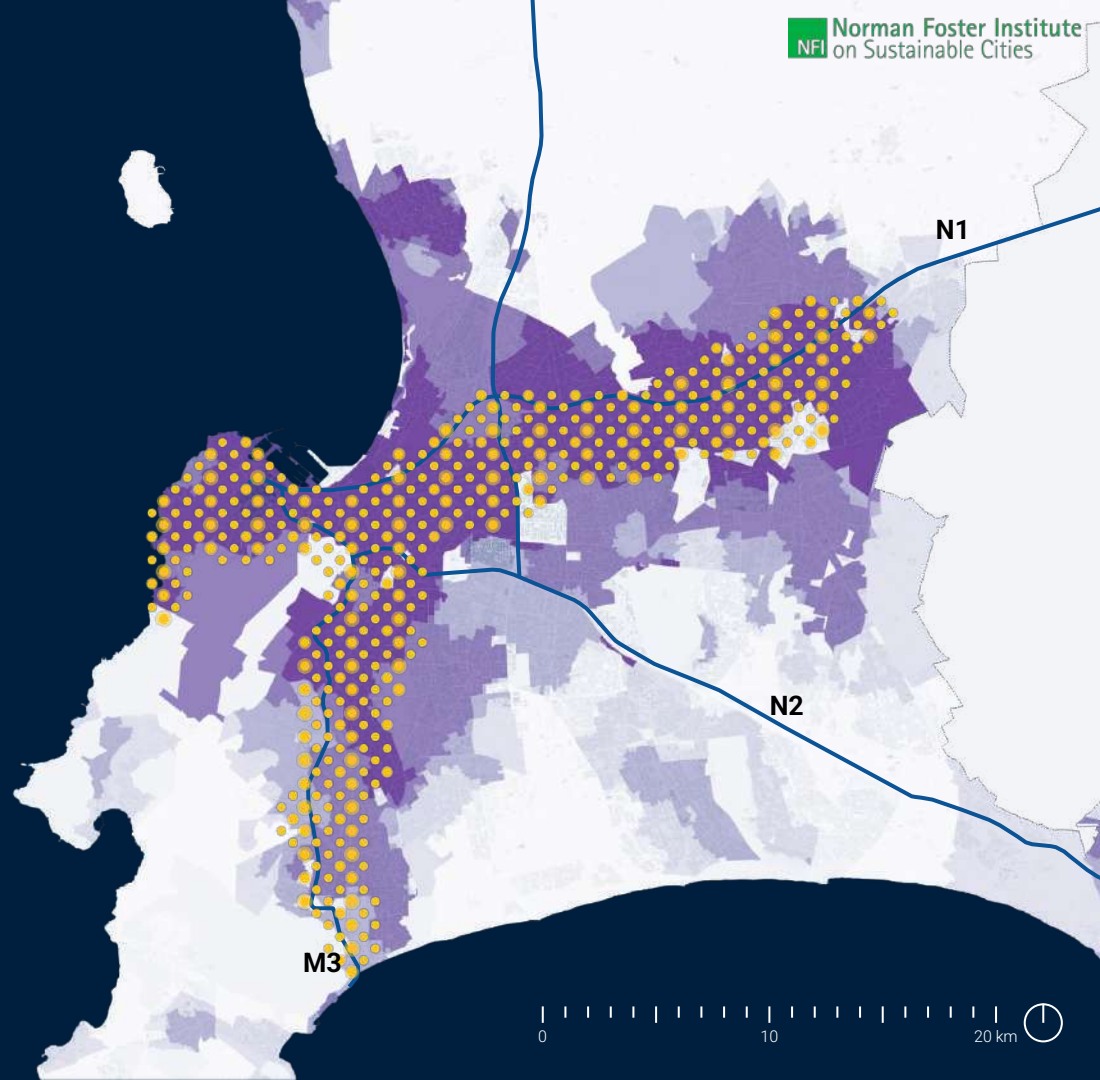
Employment Density (2011)

Low

High

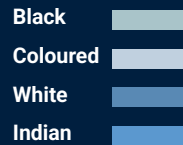


Access to Opportunities

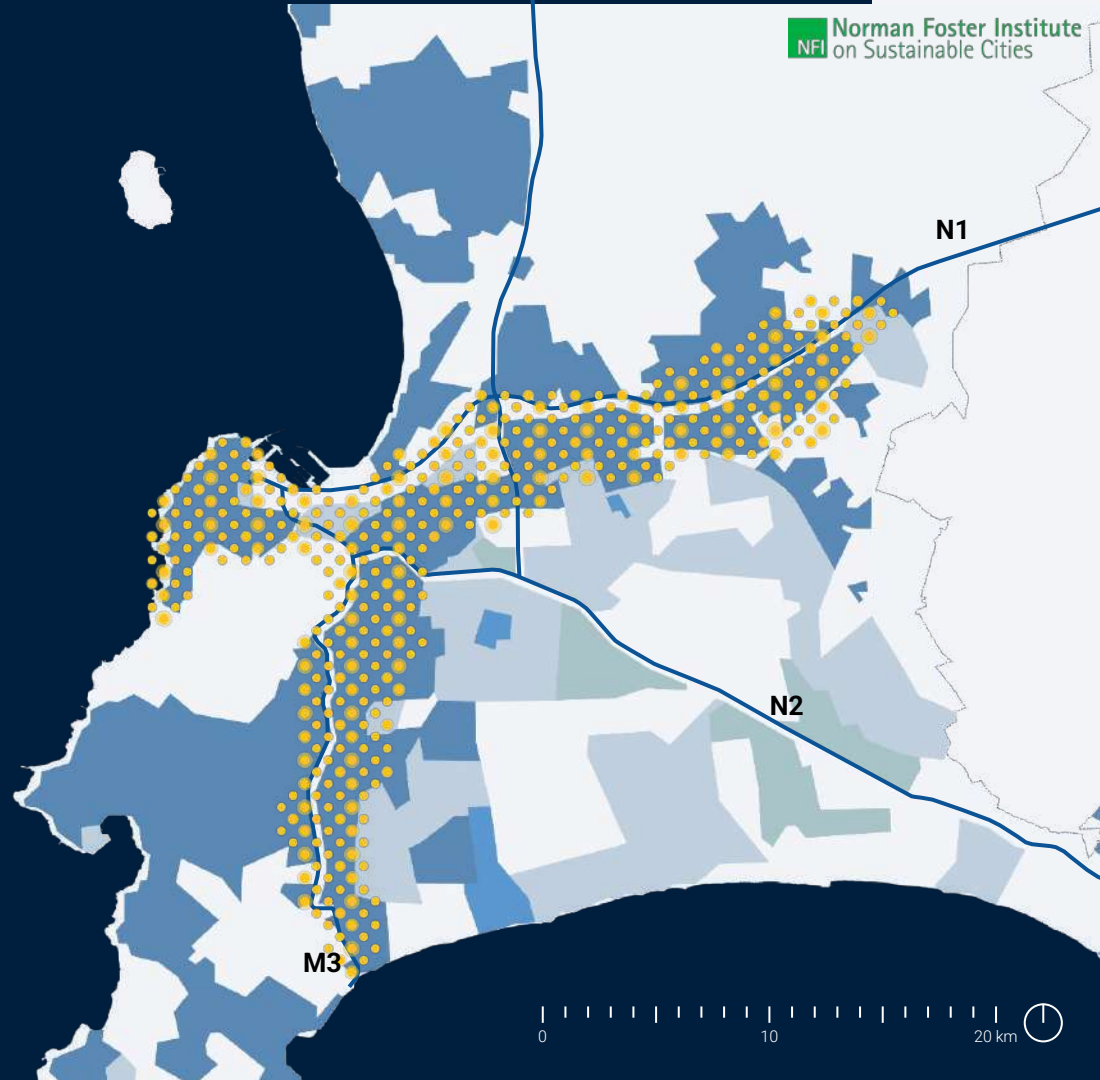


Apartheid

Group Areas Act (1948)

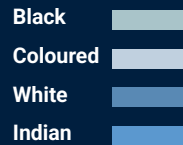


 Access to Opportunities

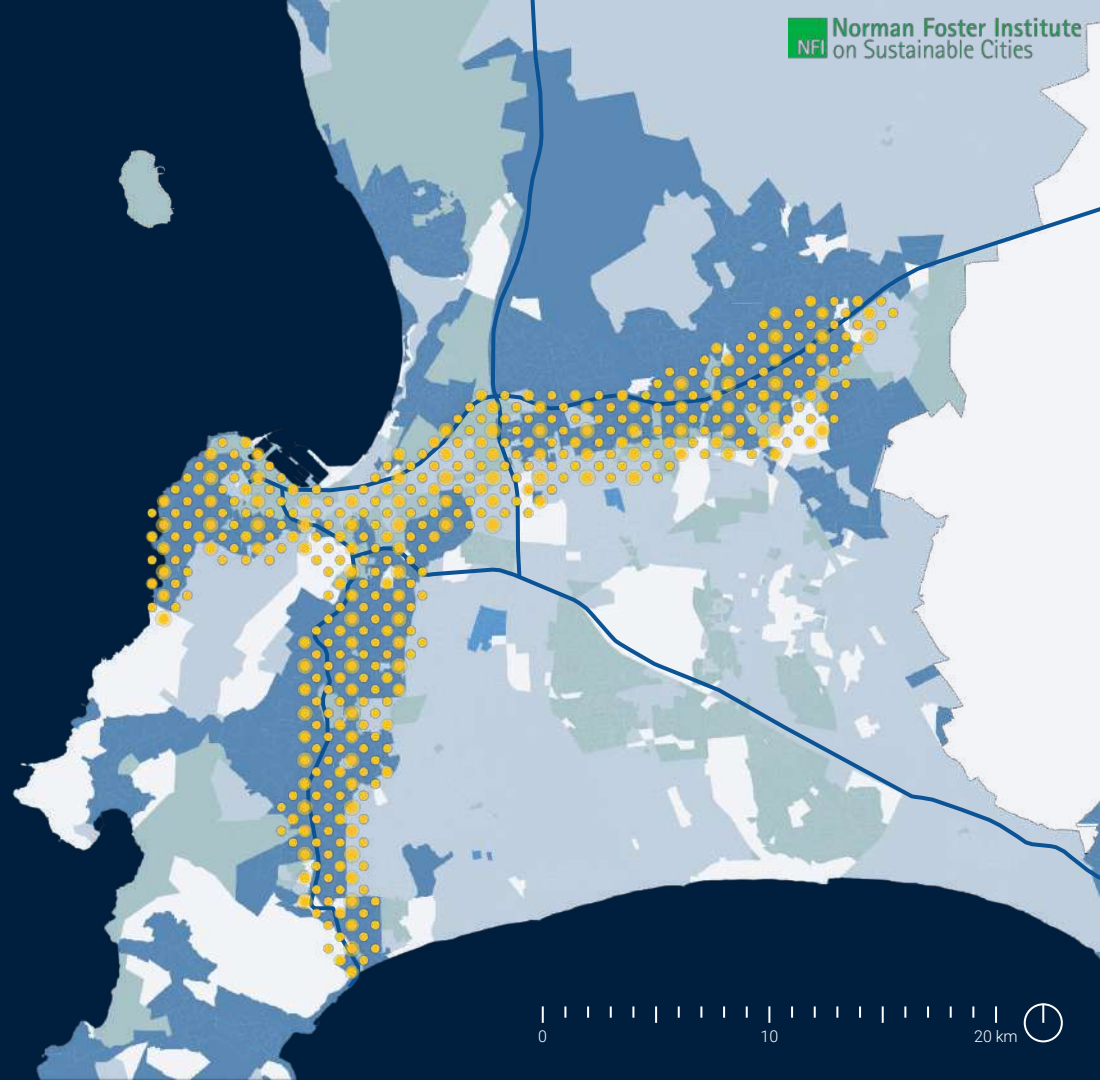


Racial Distribution

Racial Distribution (2011)



 Access to Opportunities



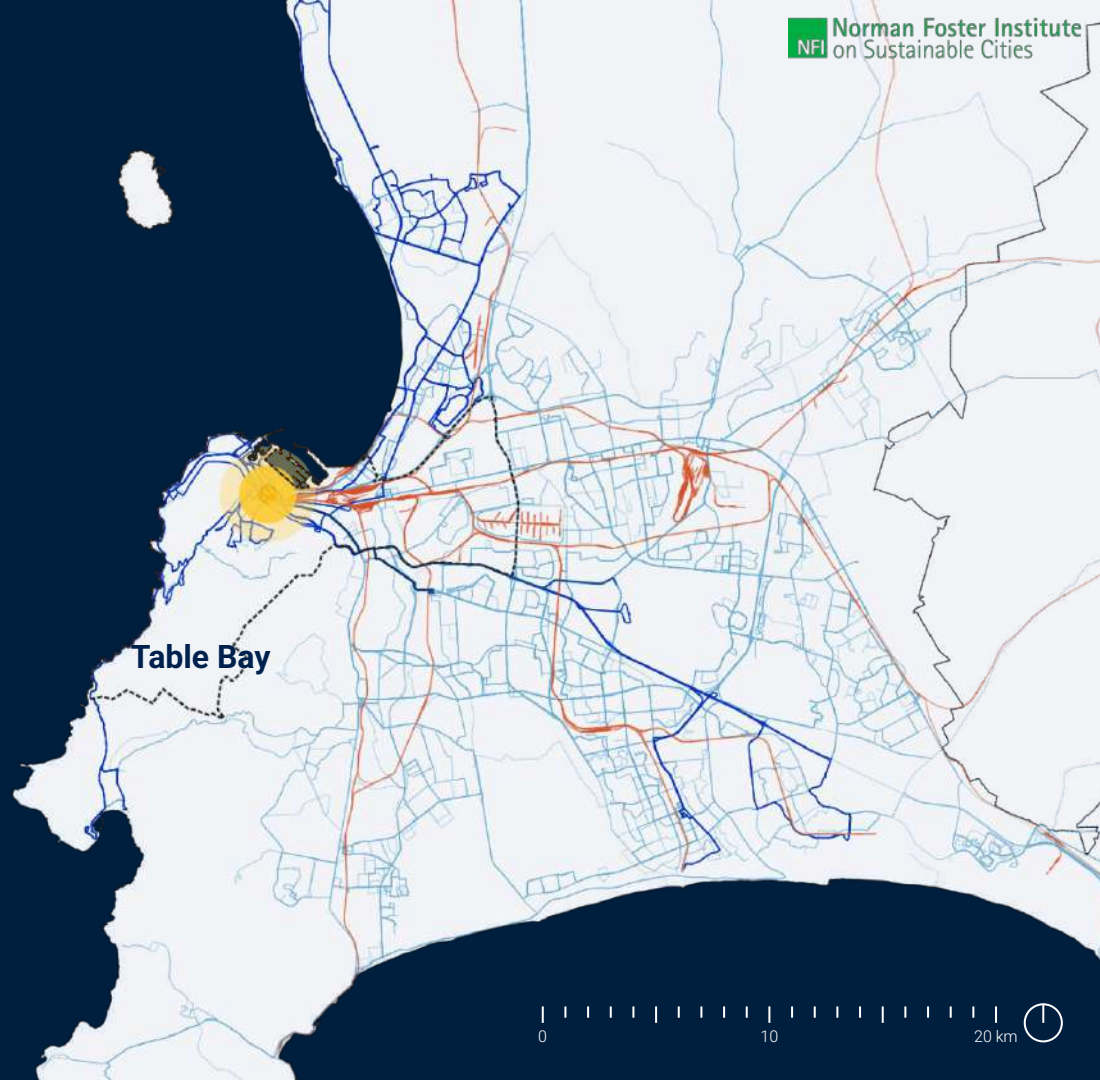
Mobility is Key

5.1M Total Population
1M Daily Commuters
into Table Bay District

Public Transport System (2025)

Minibus Taxi 
MyCiti Bus 
Rail 

 Central Business District



Mobility is Key

73% of low income workers
use public transport

43% of income spent on transport

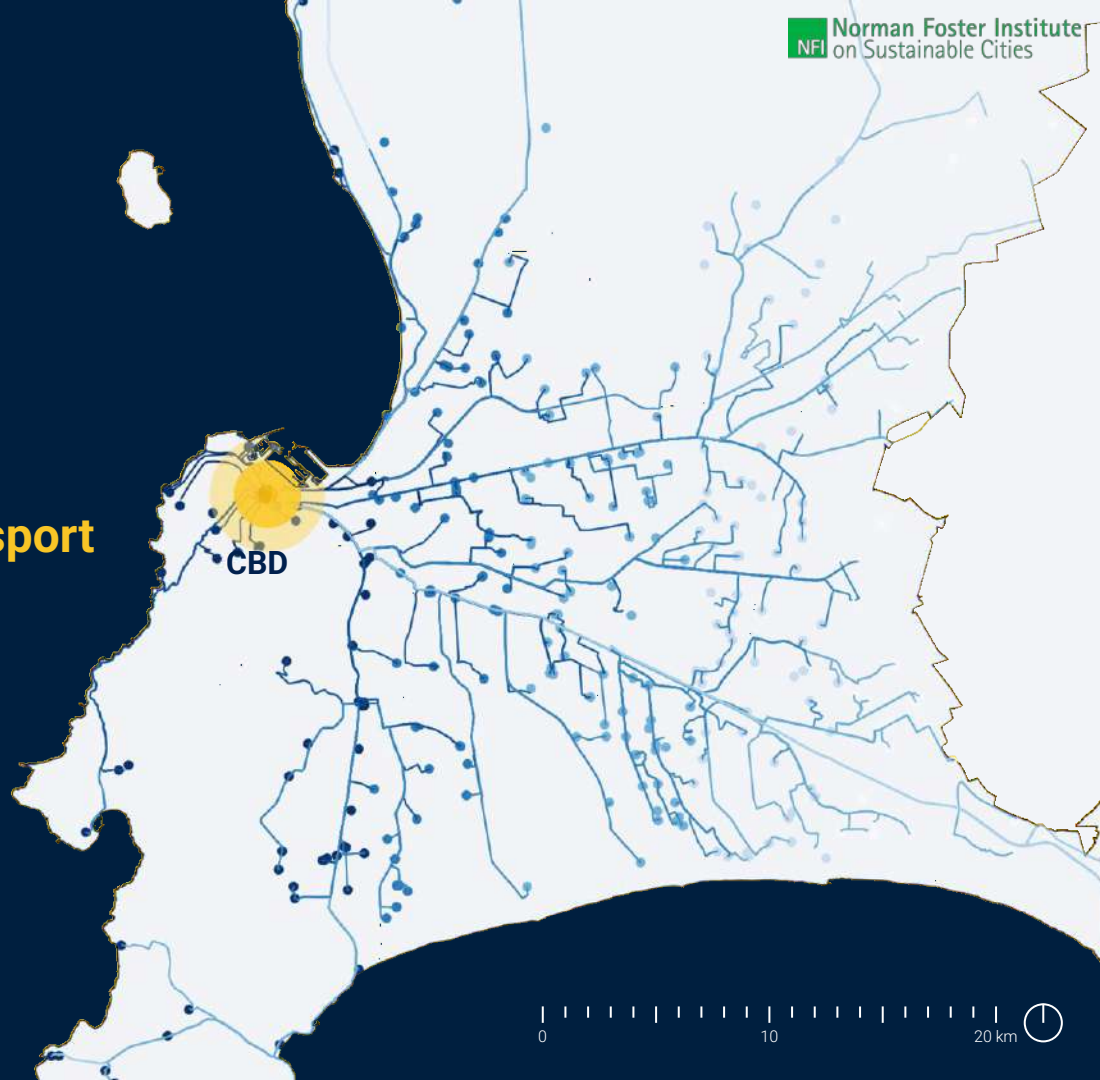
Shortest Distance of Minibus Taxi Routes
to CBD

Long

Short



Central Business District



Mobility is Key

TIME as a measure of INEQUALITY

500+ hours / year
wasted in traffic

Persona 
Minibus Taxi 

 Central Business District





4:30 am



4:30 am



5:15 am



5:15 am



6:00 am



6:00 am



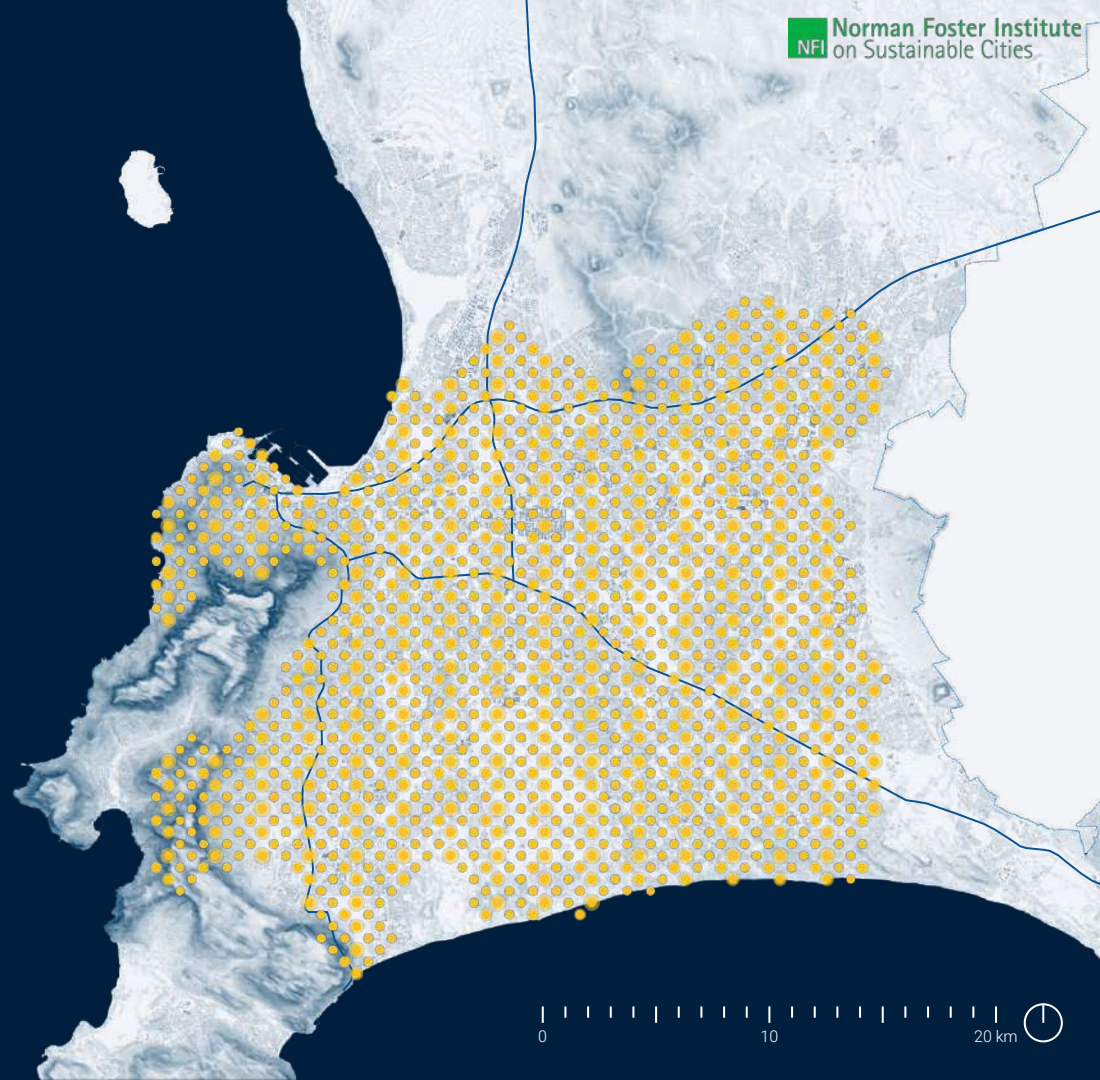
7:30 am



7:30 am

Cape Town City of Opportunity

 Access to Opportunities

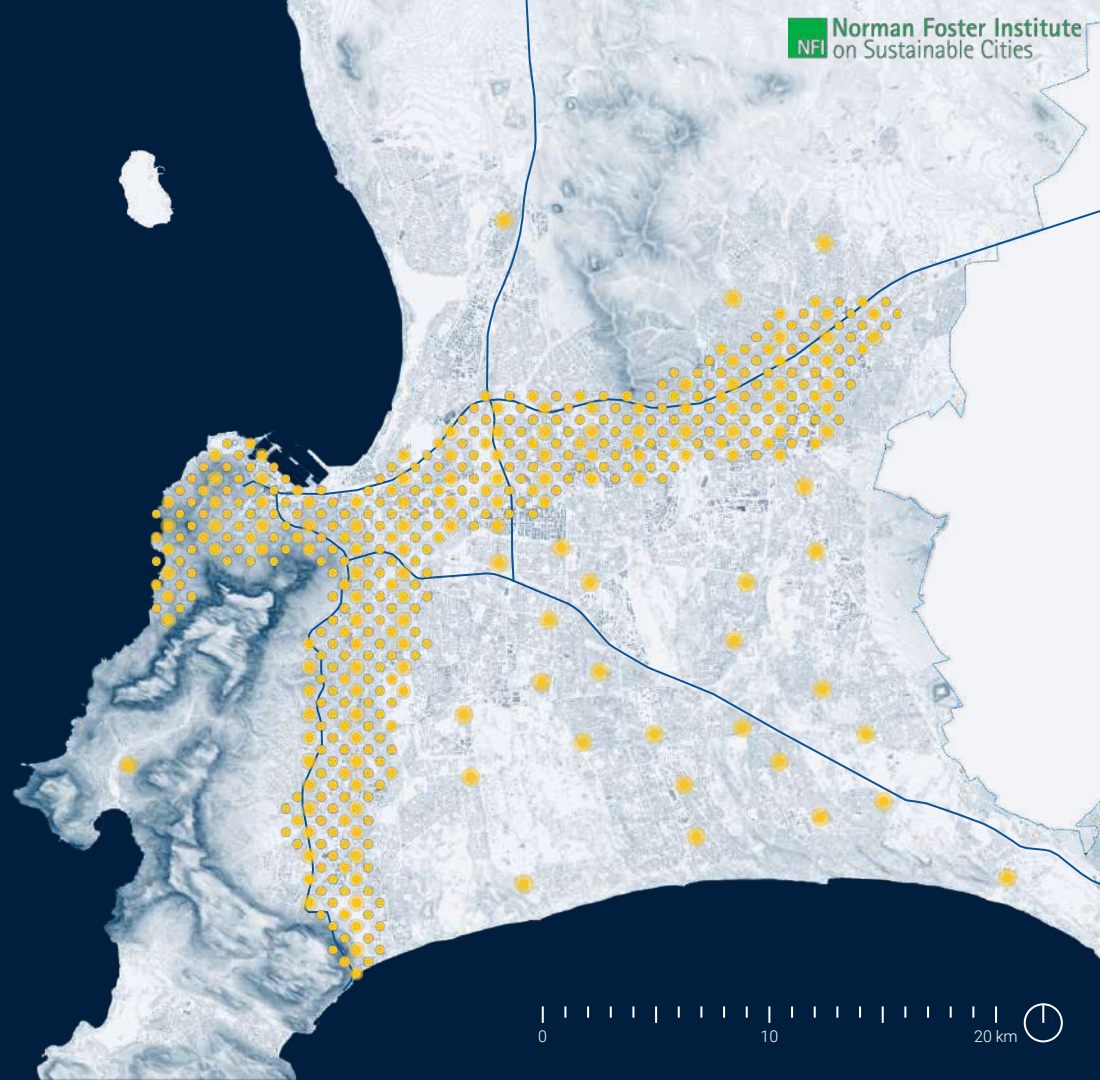


0 10 20 km 

Cape Town City of Opportunity

Opportunities for everyone,
everywhere

- Interventions
- ▣ Access to Opportunities

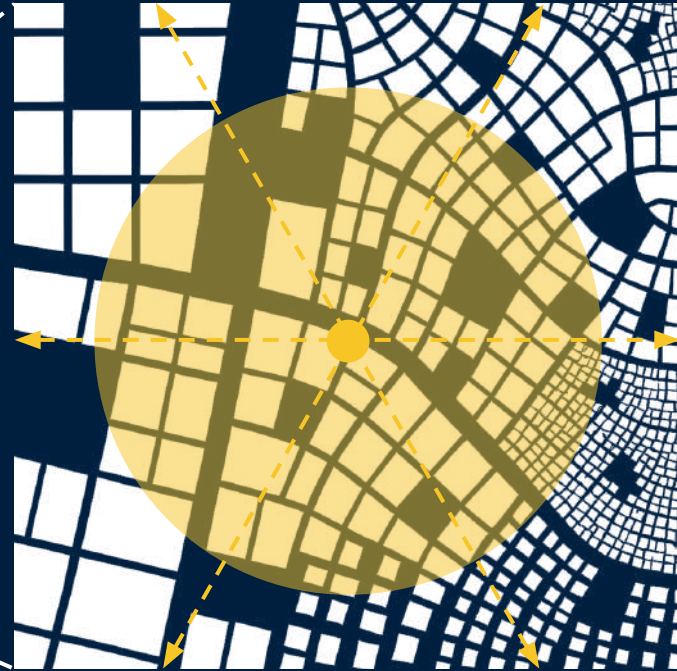


Interventions Operating at Two Scales



Metropolitan Scale

_Site Selection
_Programming & Design



Neighbourhood Scale

_Site Selection
_Programming & Design

Interventions Operating at Two Scales



Metropolitan Scale

_Site Selection
_Programming & Design



Neighbourhood Scale

_Site Selection
_Programming & Design

Interventions Operating at Two Scales

Metropolitan Scale

Transport Hubs



Thresholds



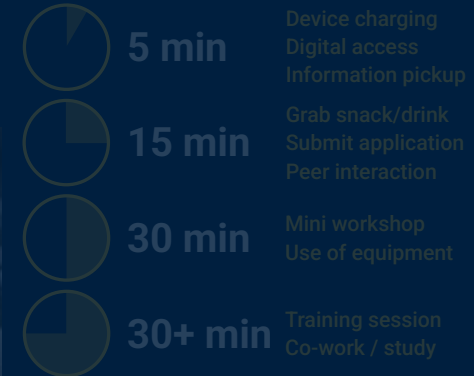
Common Programs

ENERGY
LIGHTING
CONNECTIVITY / WIFI
WATER
CULTURE /
ART SPACES

Common Standards

Standard of
Design
IDENTITY
COLOUR

Standard of
Human Experience



Site Selection

Programming & Design

Interventions Operating at Two Scales

Metropolitan Scale

Transport Hubs

Thresholds

Common Programs

Common Standards



ENERGY
LIGHTING
CONNECTIVITY / WIFI
WATER
CULTURE /
ART SPACES

Standard of
Design
IDENTITY
COLOUR

Standard of
Human Experience



5 min

Device charging
Digital access
Information pickup



15 min

Grab snack/drink
Submit application
Peer interaction



30 min

Mini workshop
Use of equipment



30+ min

Training session
Co-work / study



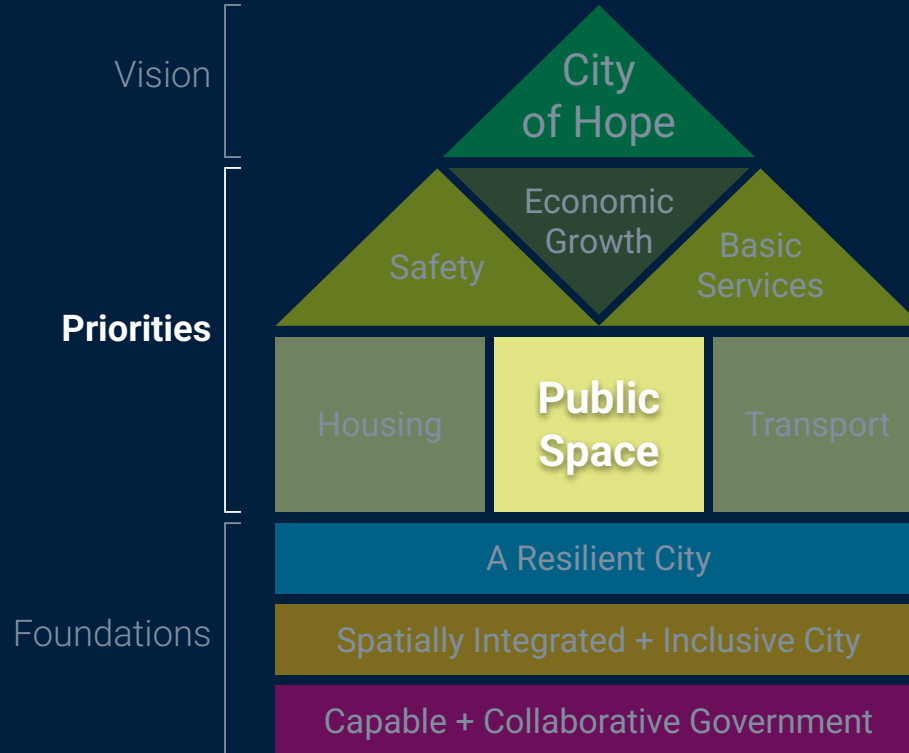
Site Selection

Programming & Design

Interventions Operating at Two Scales

Metropolitan Scale

**Strategic Alignment
with Cape Town's Policies**



Interventions Operating at Two Scales

Metropolitan Scale

Transport Hubs



Thresholds



Site Selection

Common Programs

ENERGY
LIGHTING
CONNECTIVITY / WIFI
WATER
CULTURE /
REST SPACES
ART



Common Standards

Standard of
Design

IDENTITY
COLOUR

Standard of

Human Experience



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Programming & Design

Interventions Operating at Two Scales

Metropolitan Scale

Transport Hubs



Thresholds



Site Selection

Common Programs

ENERGY

LIGHTING

CONNECTIVITY / WIFI

WATER

CULTURE /

REST SPACES



Common Standards

Standard of
Design

IDENTITY

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Standard of
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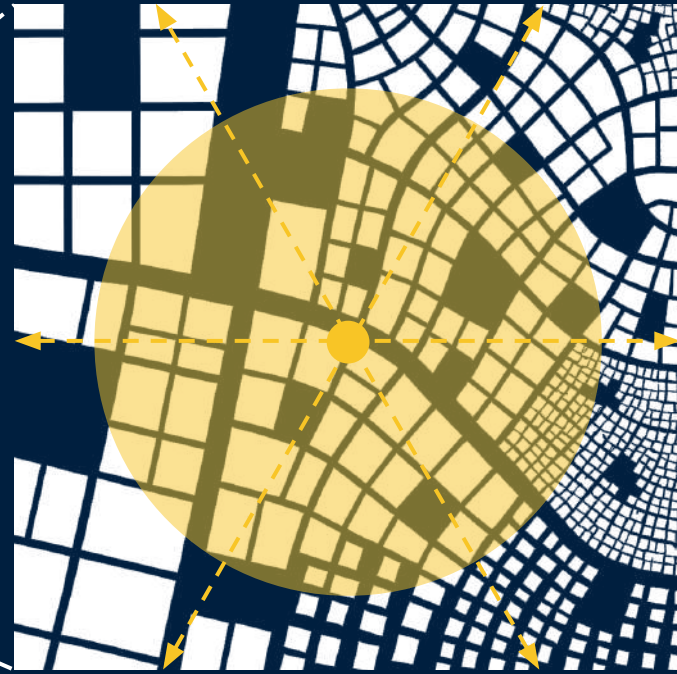
Programming & Design

Interventions Operating at Two Scales



Metropolitan Scale

_Site Selection
_Programming & Design



Neighbourhood Scale

_Site Selection
_Programming & Design

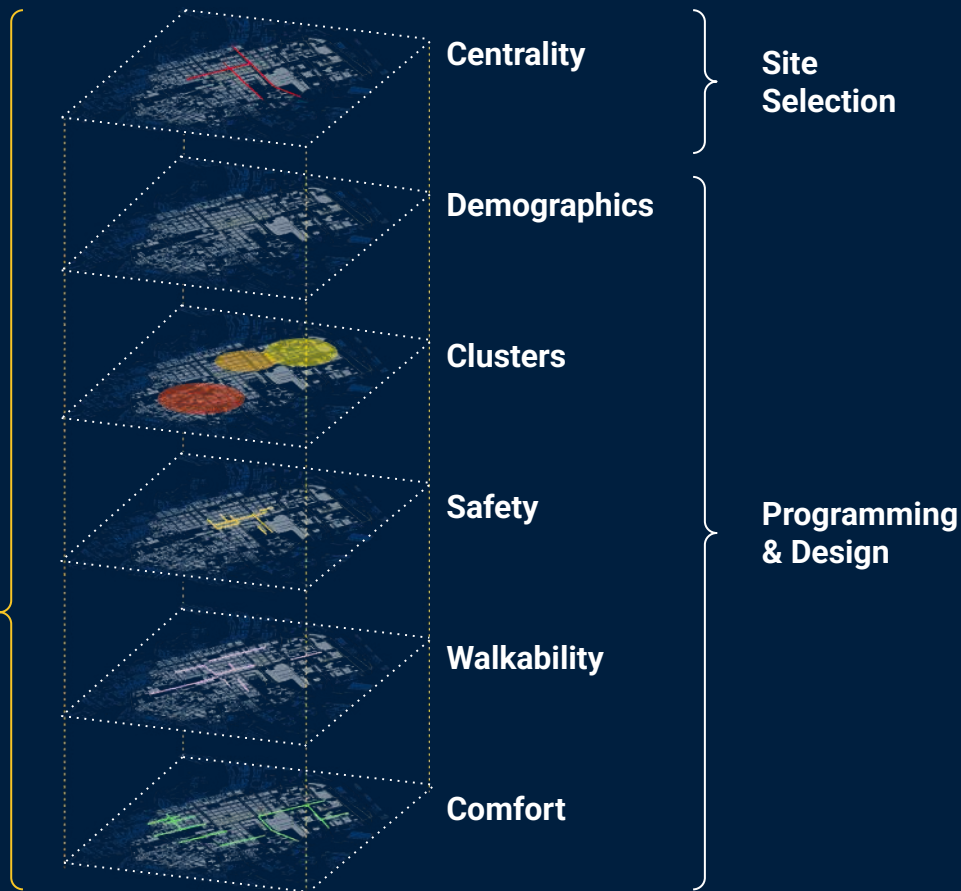
Interventions Operating at Two Scales

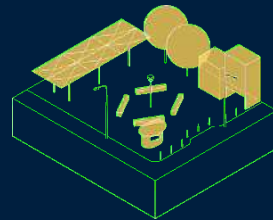
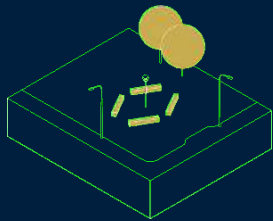
Neighbourhood Scale

Context Responsive Interventions:

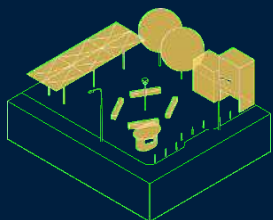
What does **Opportunity** mean
in the neighbourhood?

**Opportunity
Index**

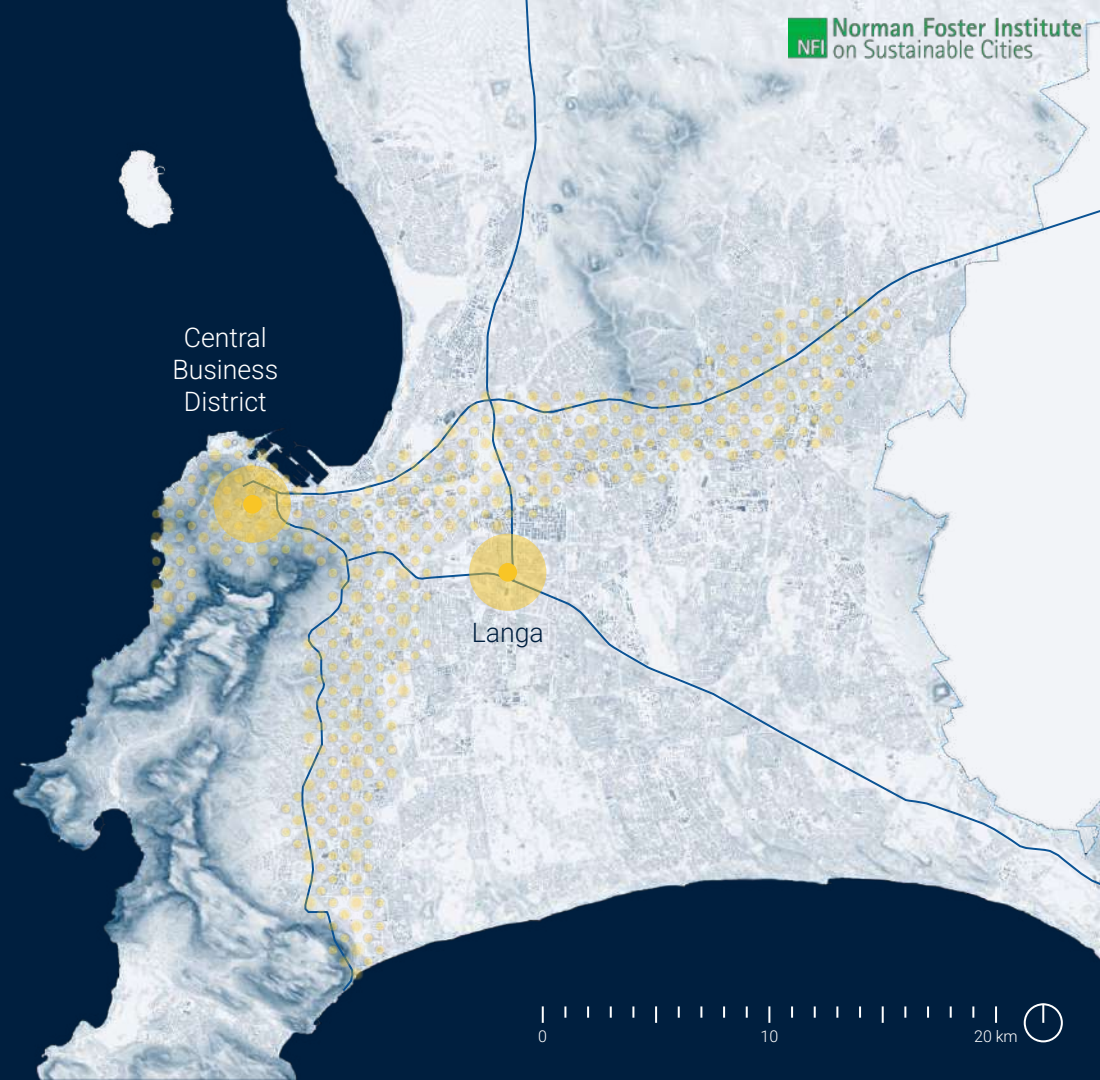




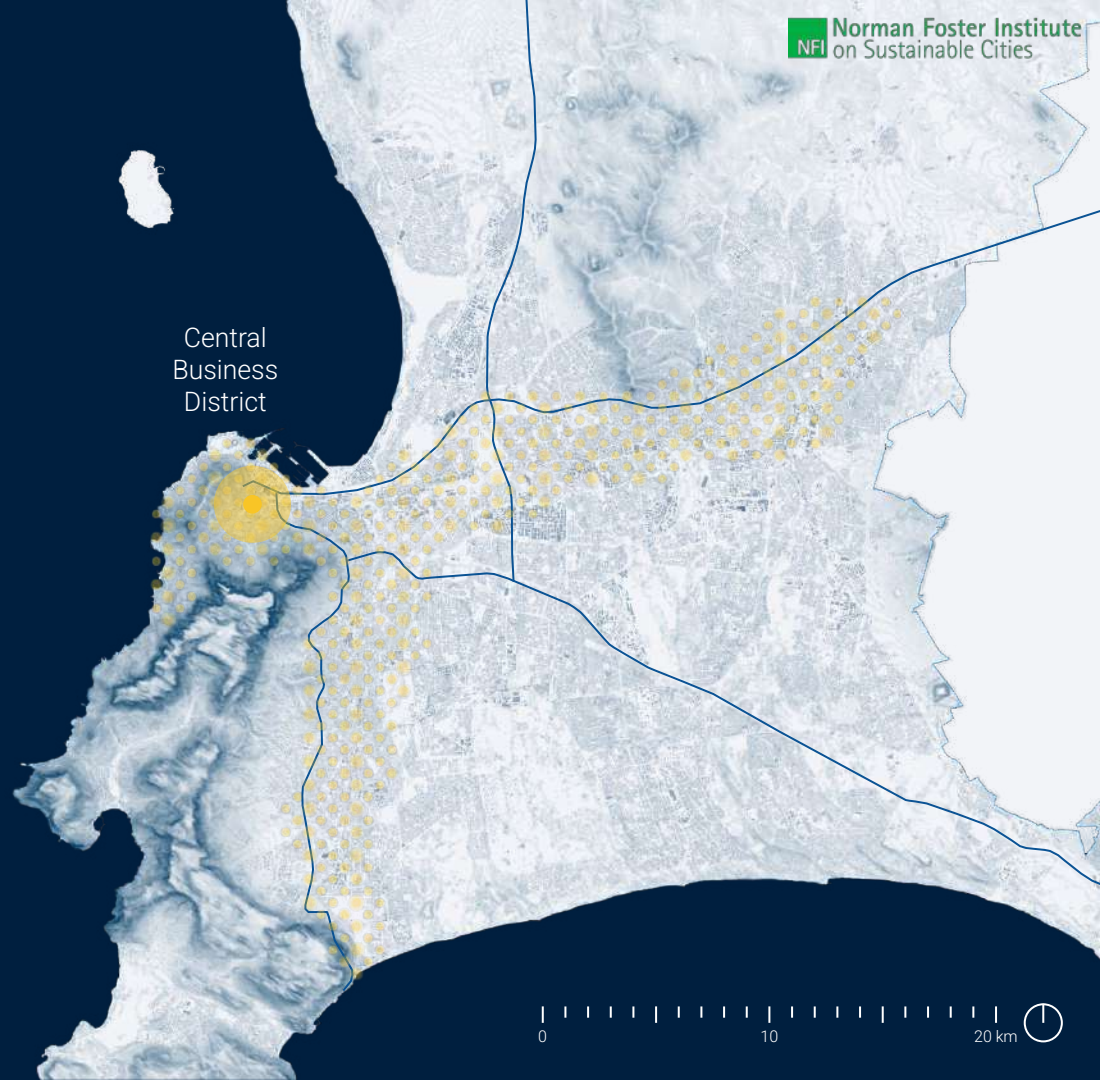
OPPORTUNITY HUBS



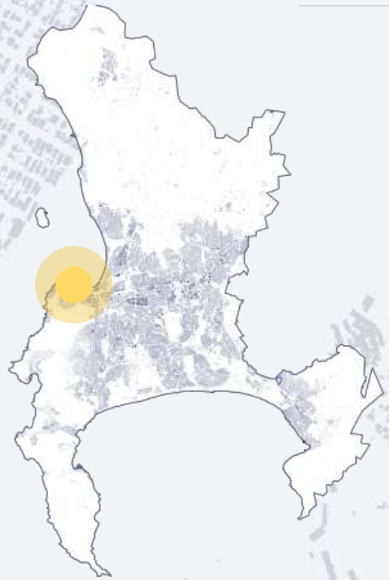
Pilot Projects



Pilot Projects



Pilot Project: Central Business District



Pilot Project: Central Business District

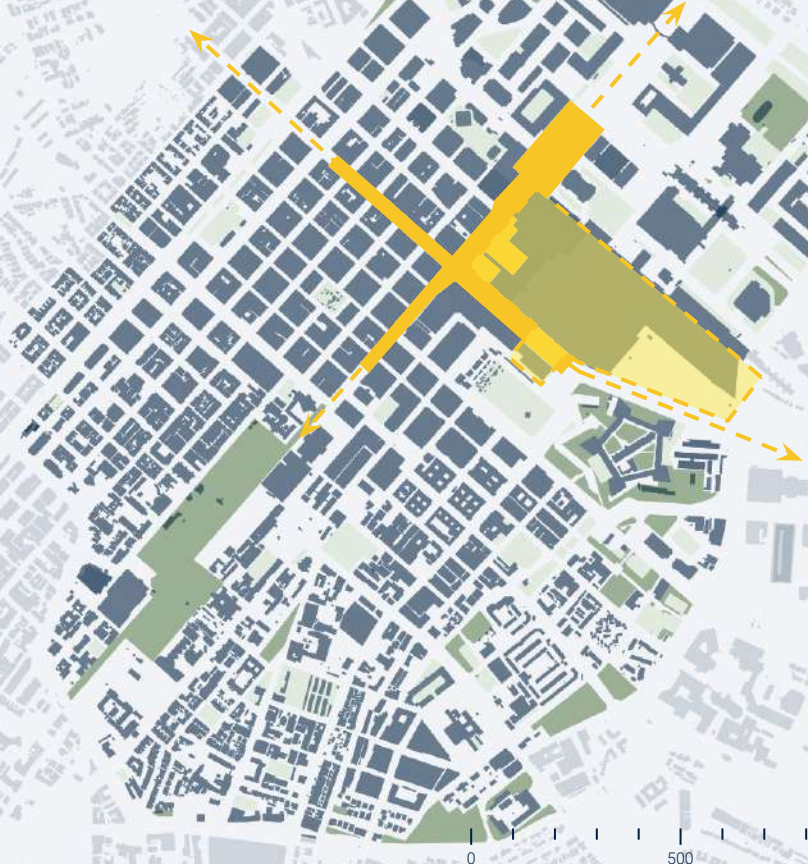
Analysis for Site Selection: Transport Hubs (Metropolitan Scale Strategy)

Cape Town Train Station Deck:
210 000 Daily Commuters



Pilot Project: Central Business District

Analysis for Site Selection: Thresholds (Metropolitan Scale Strategy)



Pilot Project: Central Business District

Analysis for Site Selection: Centralities

(Neighbourhood Scale Strategy)

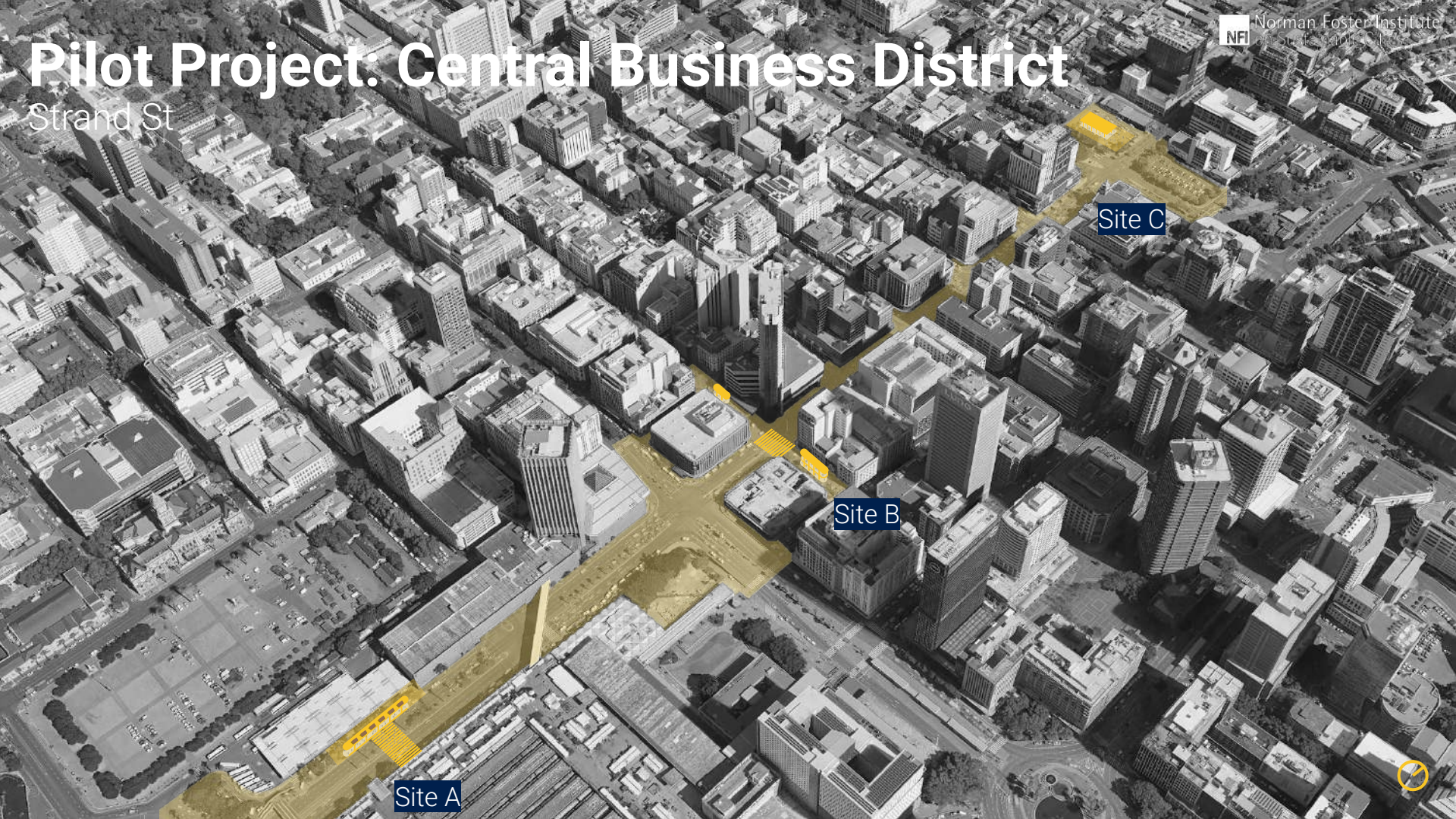
centrality_betweeness_1200
and 10000_ang_norm

- Local & Global Important Streets
- Local Important Streets
- Global Important Streets
- Buildings



Pilot Project: Central Business District

Strand St



Site C

Site B

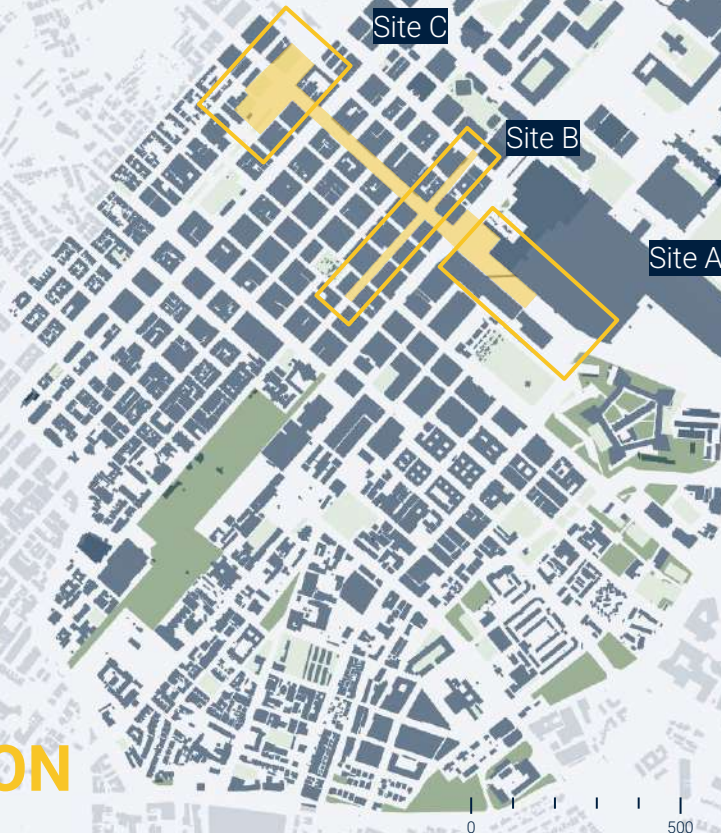
Site A



Pilot Project: Central Business District

Analysis for Programming & Design: Demographics (Neighbourhood Scale Strategy)

7 000 Residents
210 000 Daily Commuters
WORKFORCE
DIVERSE DAYTIME POPULATION



Pilot Project: Central Business District

Analysis for Programming & Design: Land Use Clusters

(Neighbourhood Scale Strategy)



Pilot Project: Central Business District

Analysis for Programming & Design: Safety

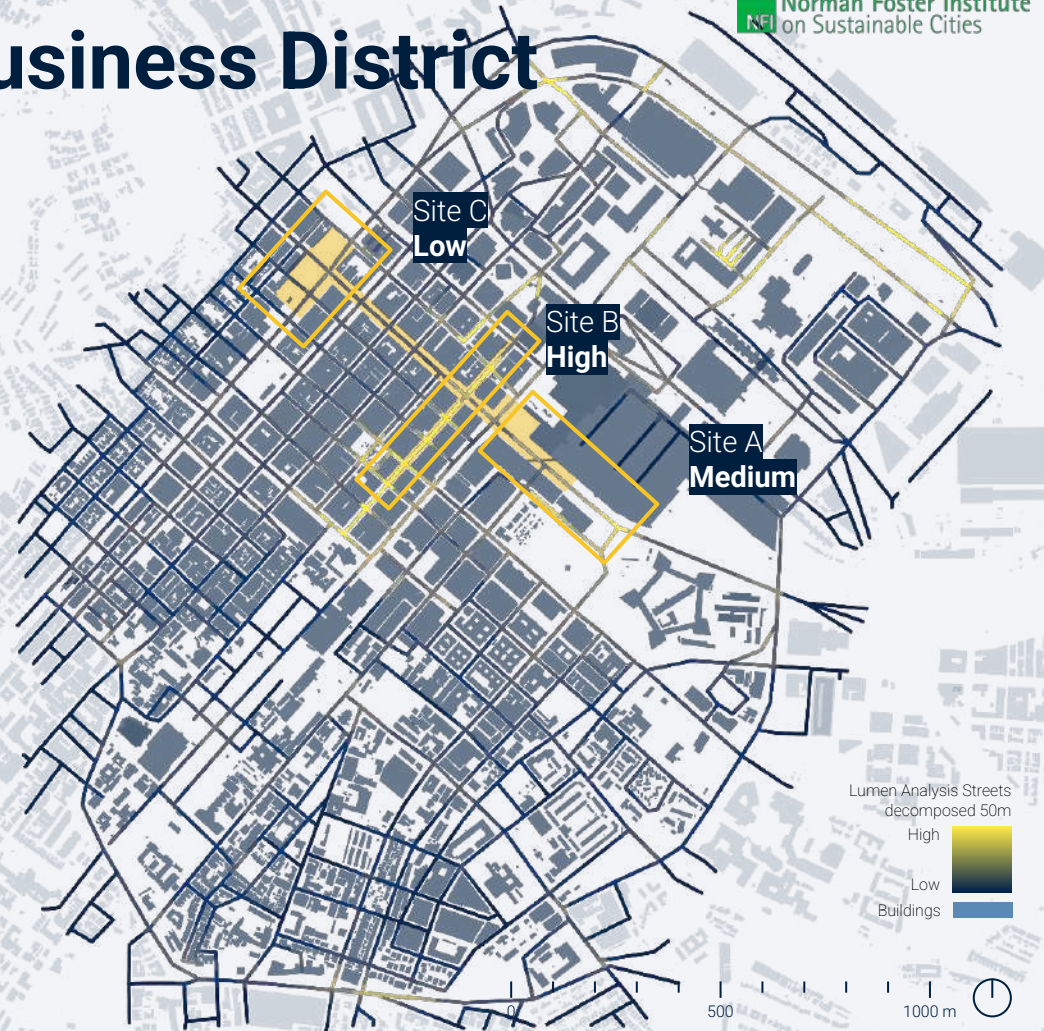
(Neighbourhood Scale Strategy)

Lumen Light Analysis

Visibility Analysis

Presence of Homeless

Porosity of Storefronts



Pilot Project: Central Business District

Analysis for Programming & Design: Walkability

(Neighbourhood Scale Strategy)

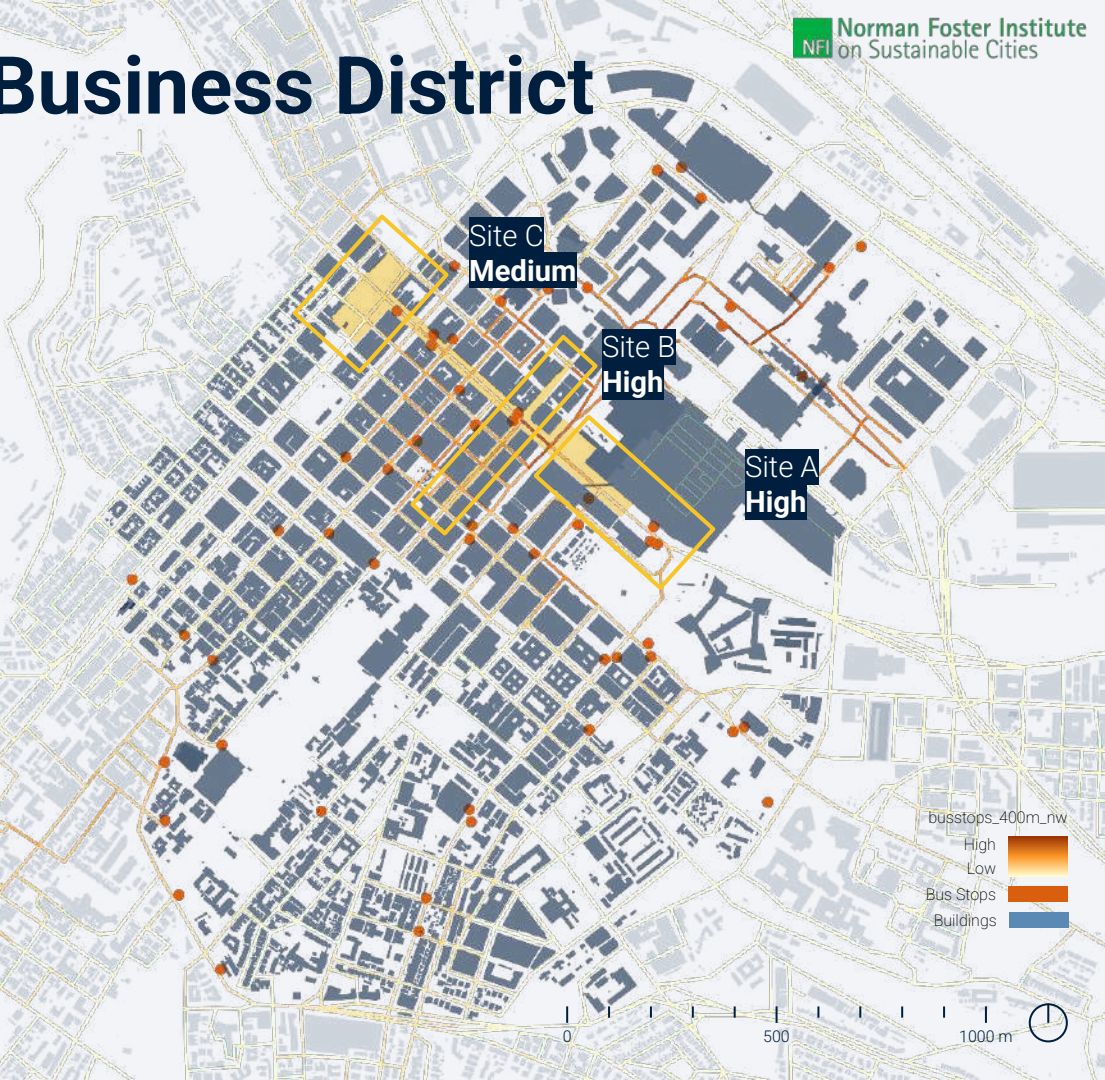
Access to Bus Network

Access to Rail Network

Access to Mini Bus Taxis

Points of Interest Analysis

Pedestrian/ Vehicle Conflict



Pilot Project: Central Business District

Analysis for Programming & Design: Comfort

(Neighbourhood Scale Strategy)

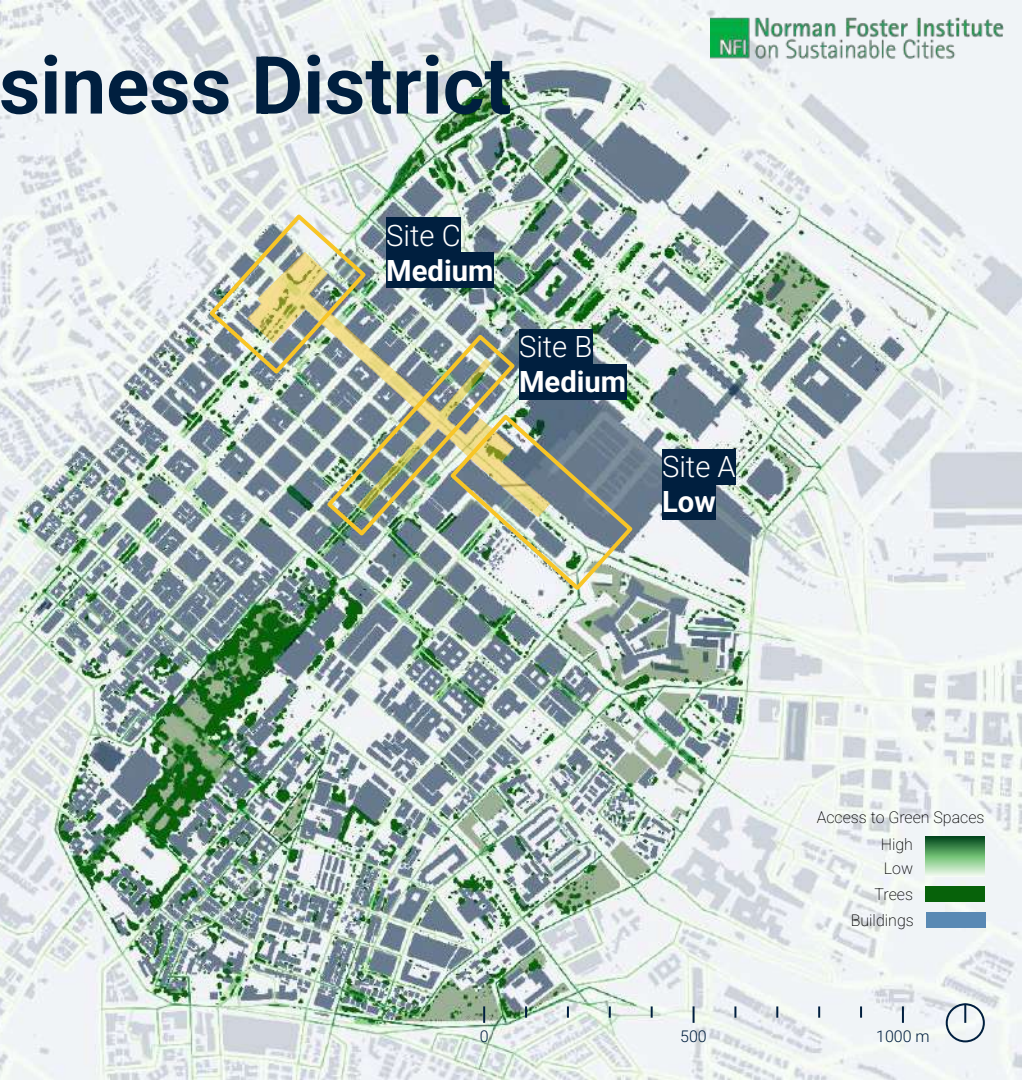
Access to Green Space

Percentage of Tree Coverage

Sky View Factor Analysis

Shadow Analysis

Presence of Urban Furniture



Pilot Project: Central Business District

Site A: Strand St + Station Deck



Buitengracht

St. Georges Mall

Station Deck

Demographics

Land Use Clusters

Comfort

Safety

Walkability

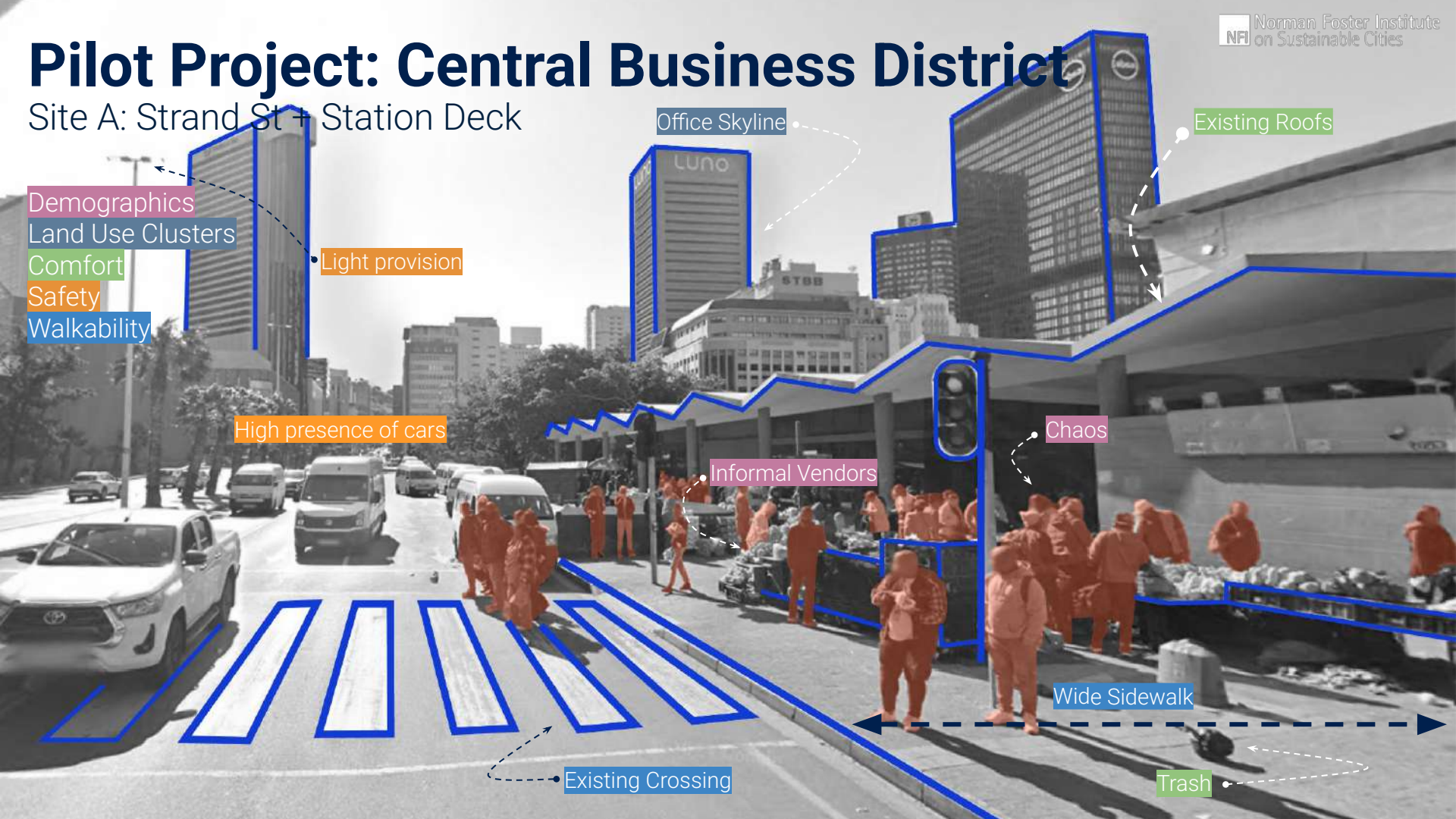
High presence of cars

Informal Vendors

Chaos

- Existing Crossing

Trash



Pilot Project: Central Business District

Site A: Strand St + Station Deck

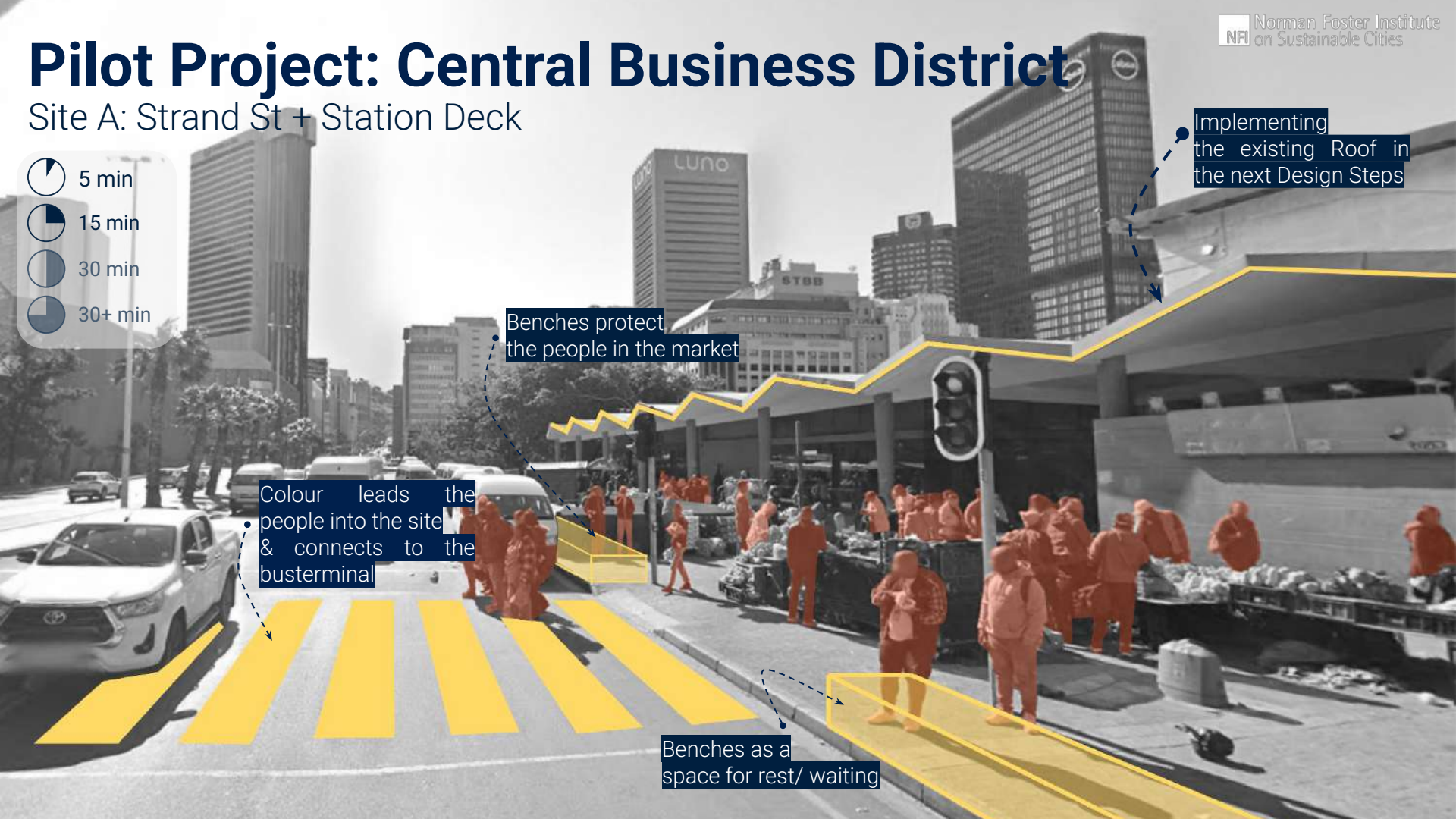


Implementing
the existing Roof in
the next Design Steps

Benches protect
the people in the market

Colour leads the
people into the site
& connects to the
busterminal

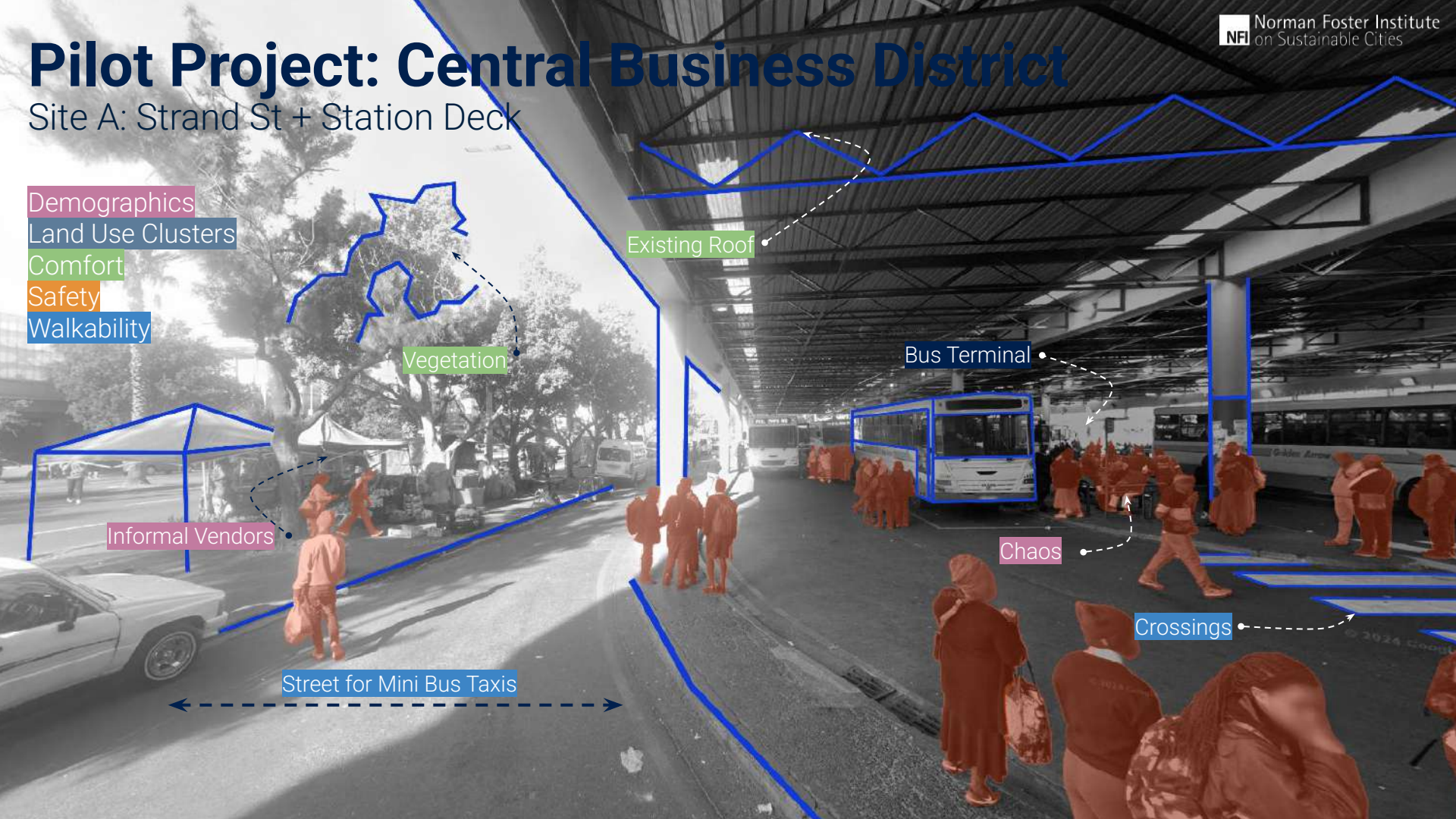
Benches as a
space for rest/ waiting



Pilot Project: Central Business District

Site A: Strand St + Station Deck

- Demographics
- Land Use Clusters
- Comfort
- Safety
- Walkability



Existing Roof

Vegetation

Bus Terminal

Informal Vendors

Chaos

Crossings

Street for Mini Bus Taxis

 5 min 15 min

 30 min

 30+ min

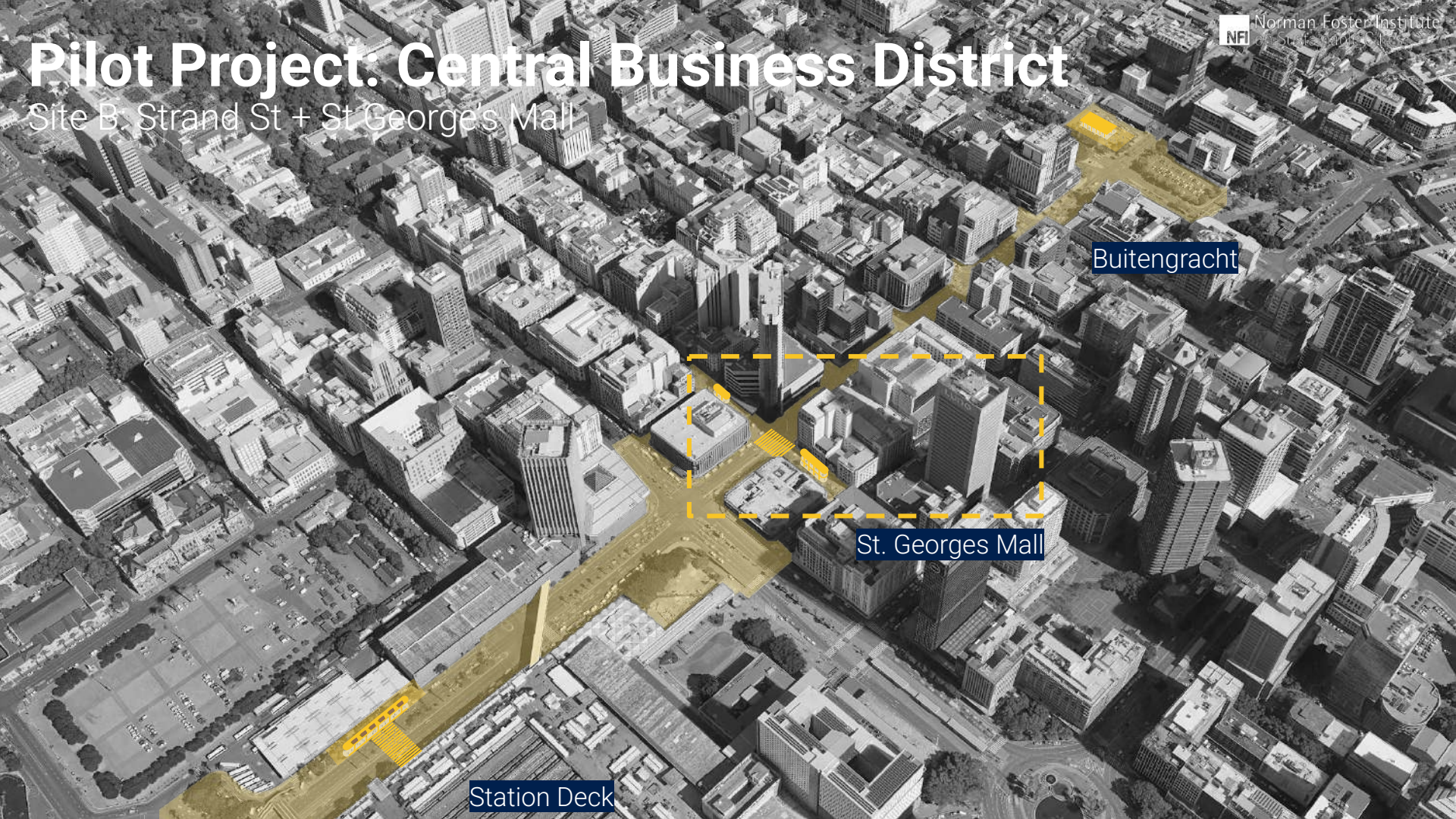
30+ min

Bus Terminal

Quick Programming Information

Pilot Project: Central Business District

Site B: Strand St + St George's Mall



Buitengracht

St. Georges Mall

Station Deck

Pilot Project: Central Business District

Site B: Strand St + St George's Mall

- Demographics
- Land Use Clusters
- Comfort
- Safety
- Walkability

Vertical Structures

Vegetation

Residential

Storefronts

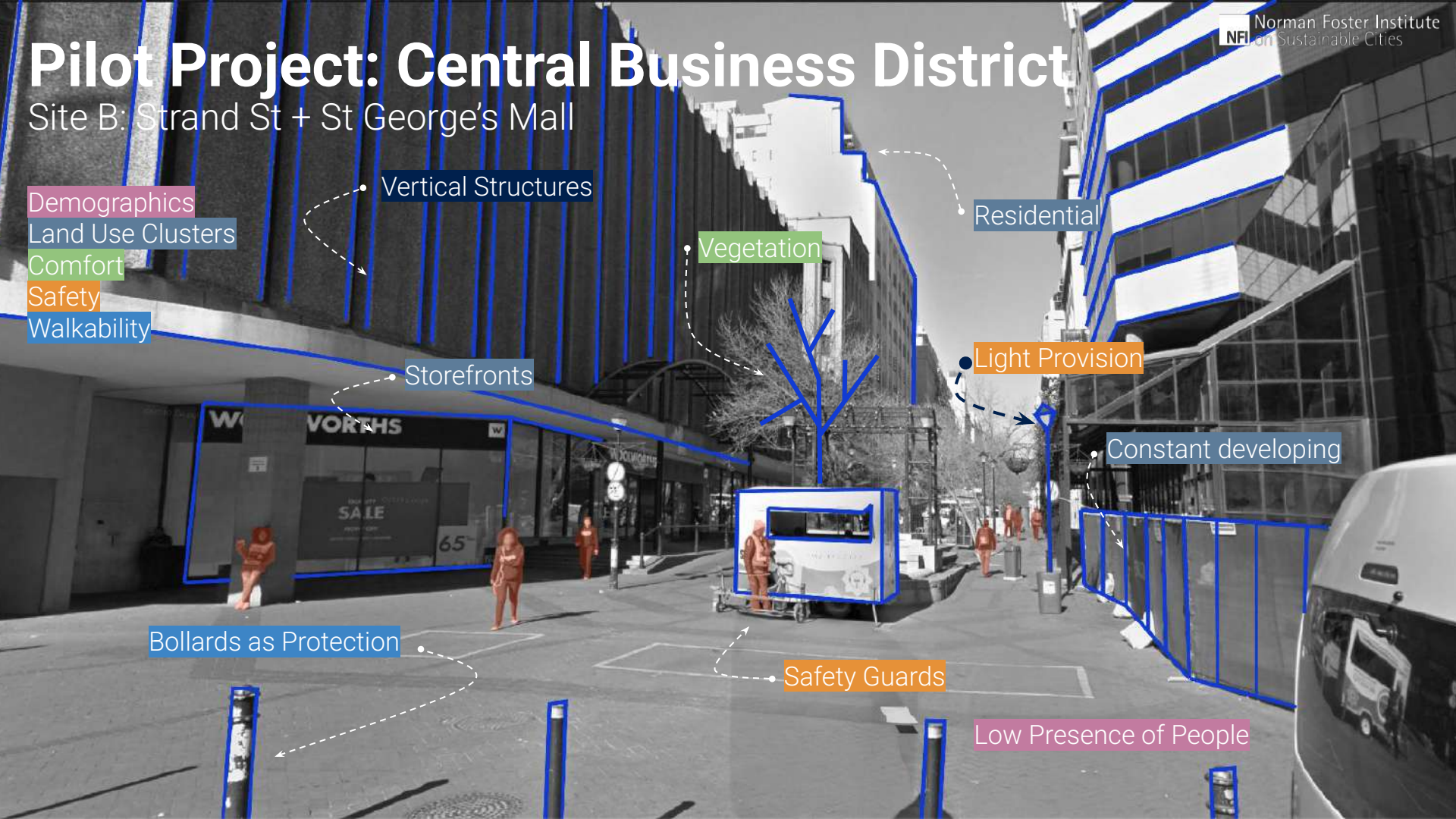
Light Provision

Constant developing

Bollards as Protection

Safety Guards

Low Presence of People



Pilot Project: Central Business District

Site B: Strand St + St George's Mall

- 5 min
- 15 min
- 30 min
- 30+ min

Colour as a Connection

Co. working hub

Trees creating comfort

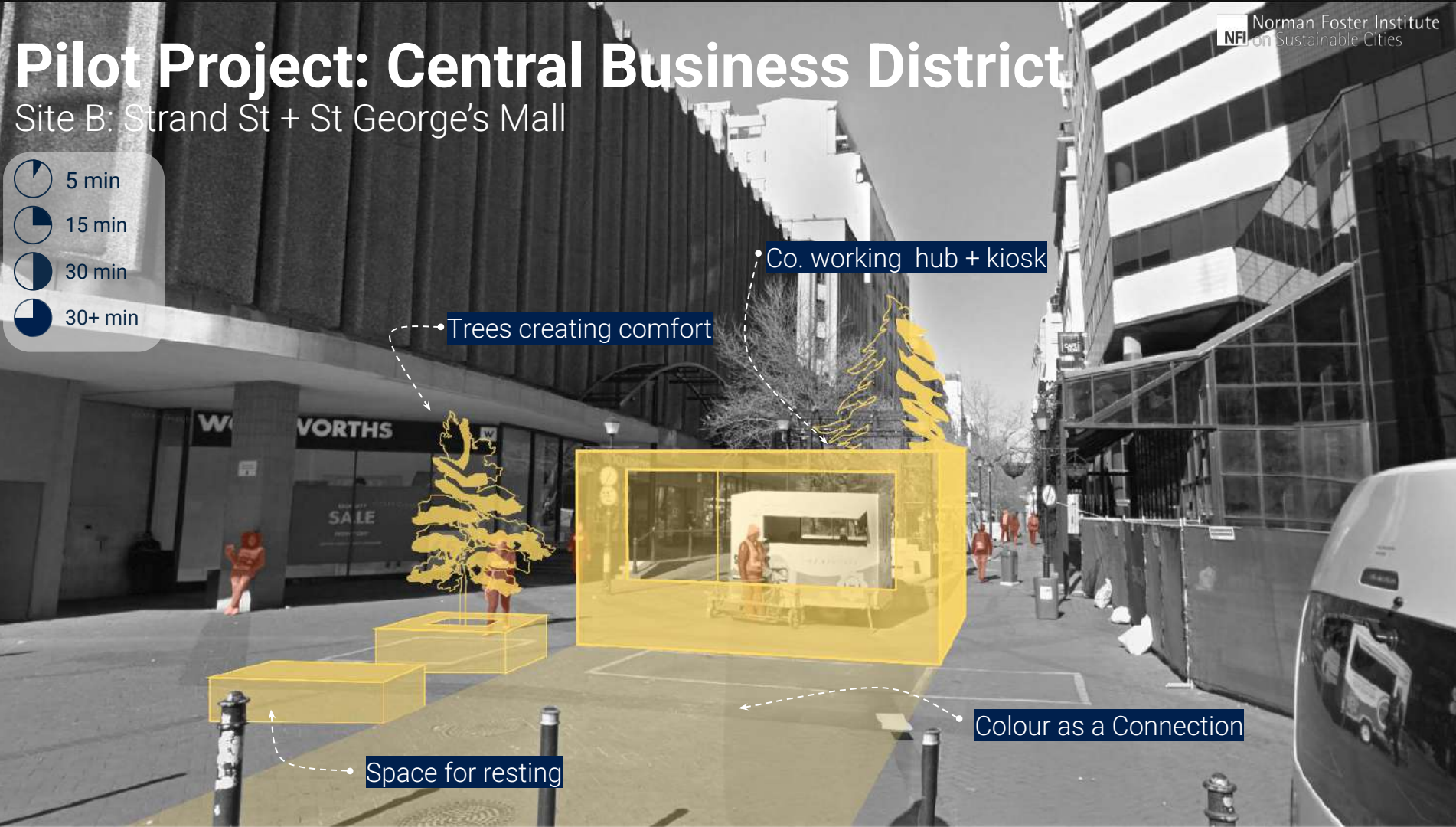
Space for resting



Pilot Project: Central Business District

Site B: Strand St + St George's Mall

- 5 min
- 15 min
- 30 min
- 30+ min



Trees creating comfort

Co. working hub + kiosk

Space for resting

Colour as a Connection

Pilot Project: Central Business District

Site C: Strand St + Buitengracht St



Buitengracht

St. Georges Mall

Station Deck

Demographics
Land Use Clusters
Comfort
Safety
Walkability

Iconic mountain range

Parking space

Fence

Destroyed sidewalks

Pilot Project: Central Business District

Site C: Strand St + Buitengracht St



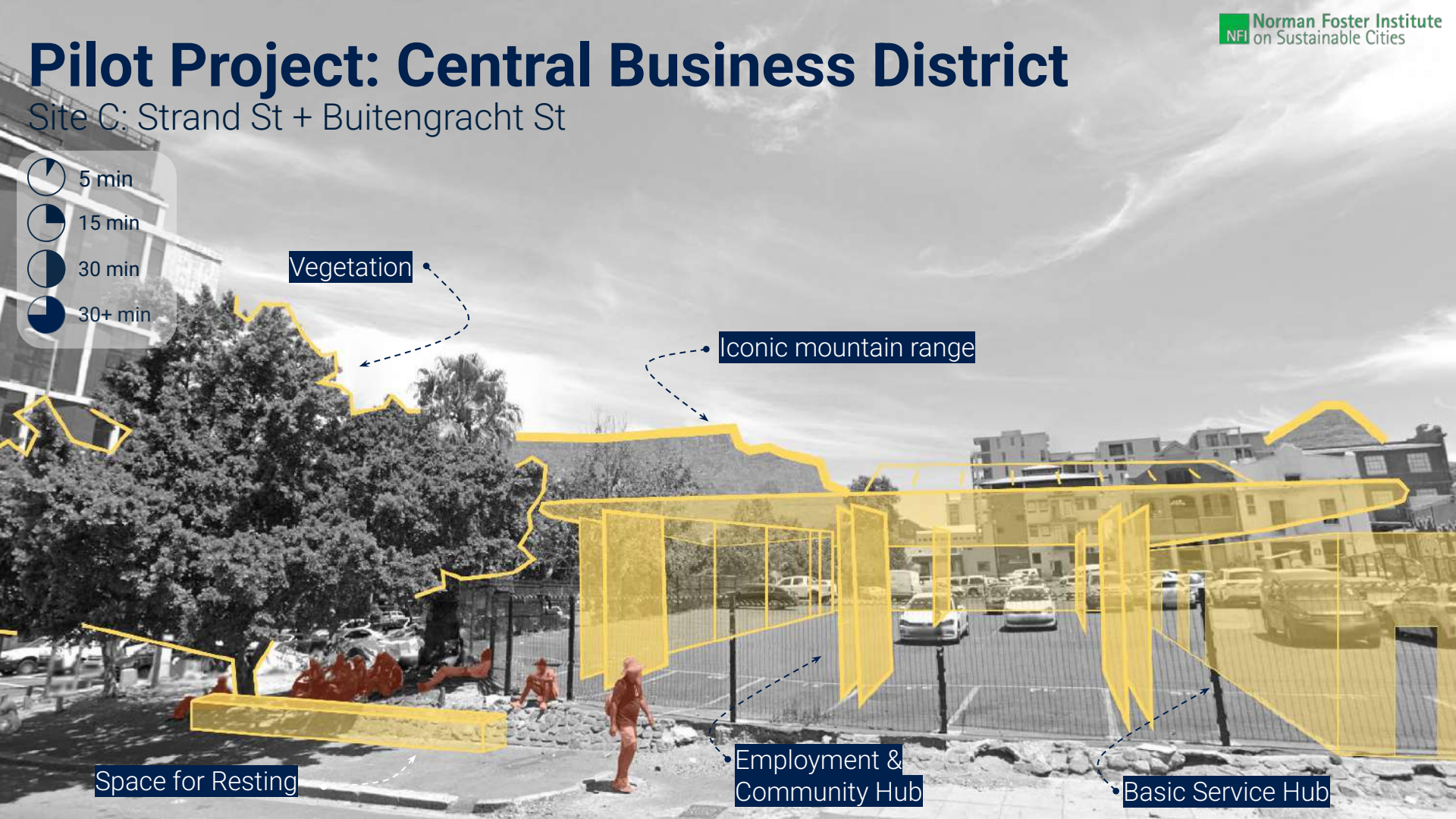
Vegetation

Iconic mountain range

Space for Resting

Employment & Community Hub

Basic Service Hub



Pilot Project: Central Business District

Site C: Strand St + Buitengracht St



Vegetation

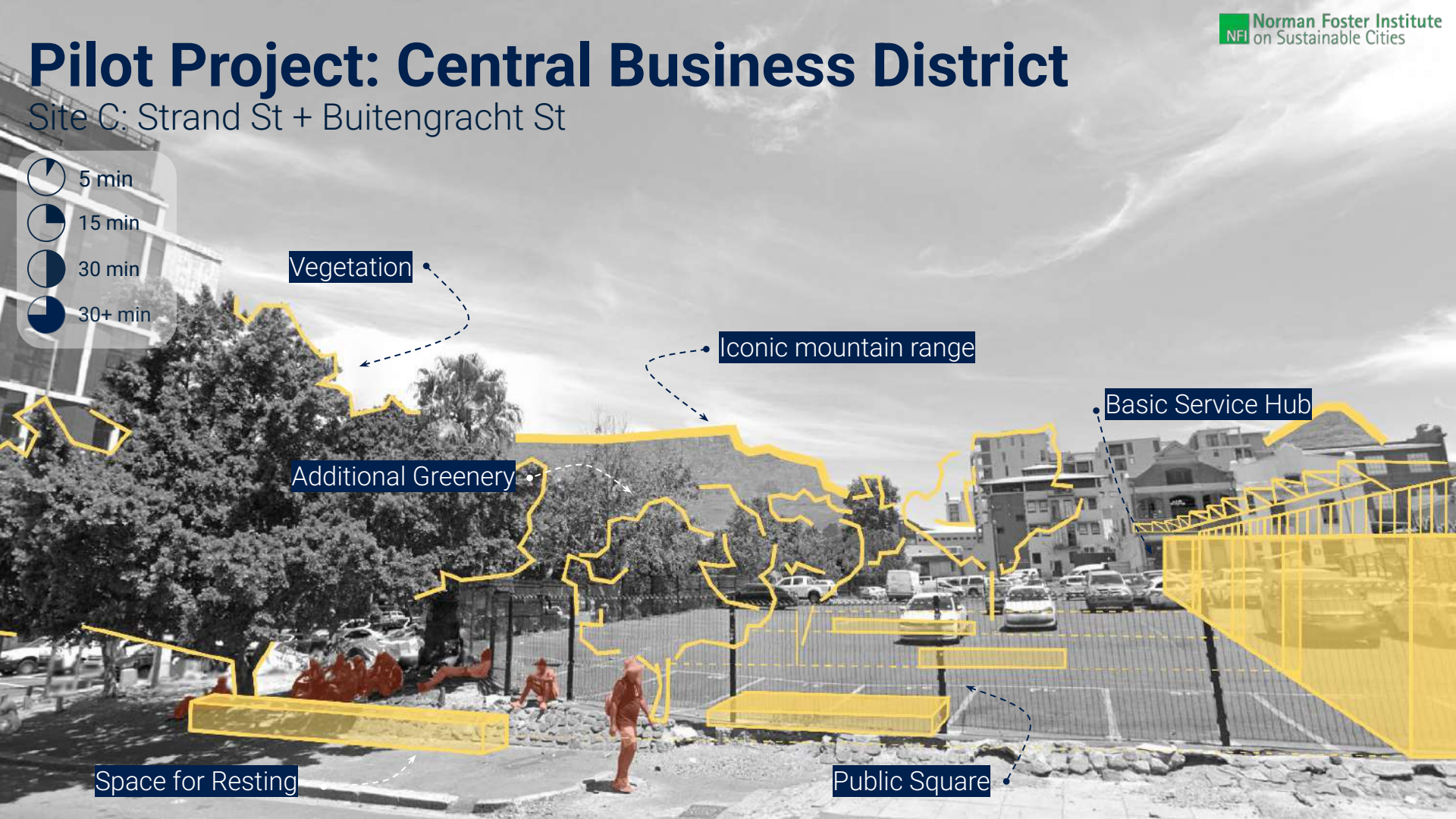
Iconic mountain range

Basic Service Hub

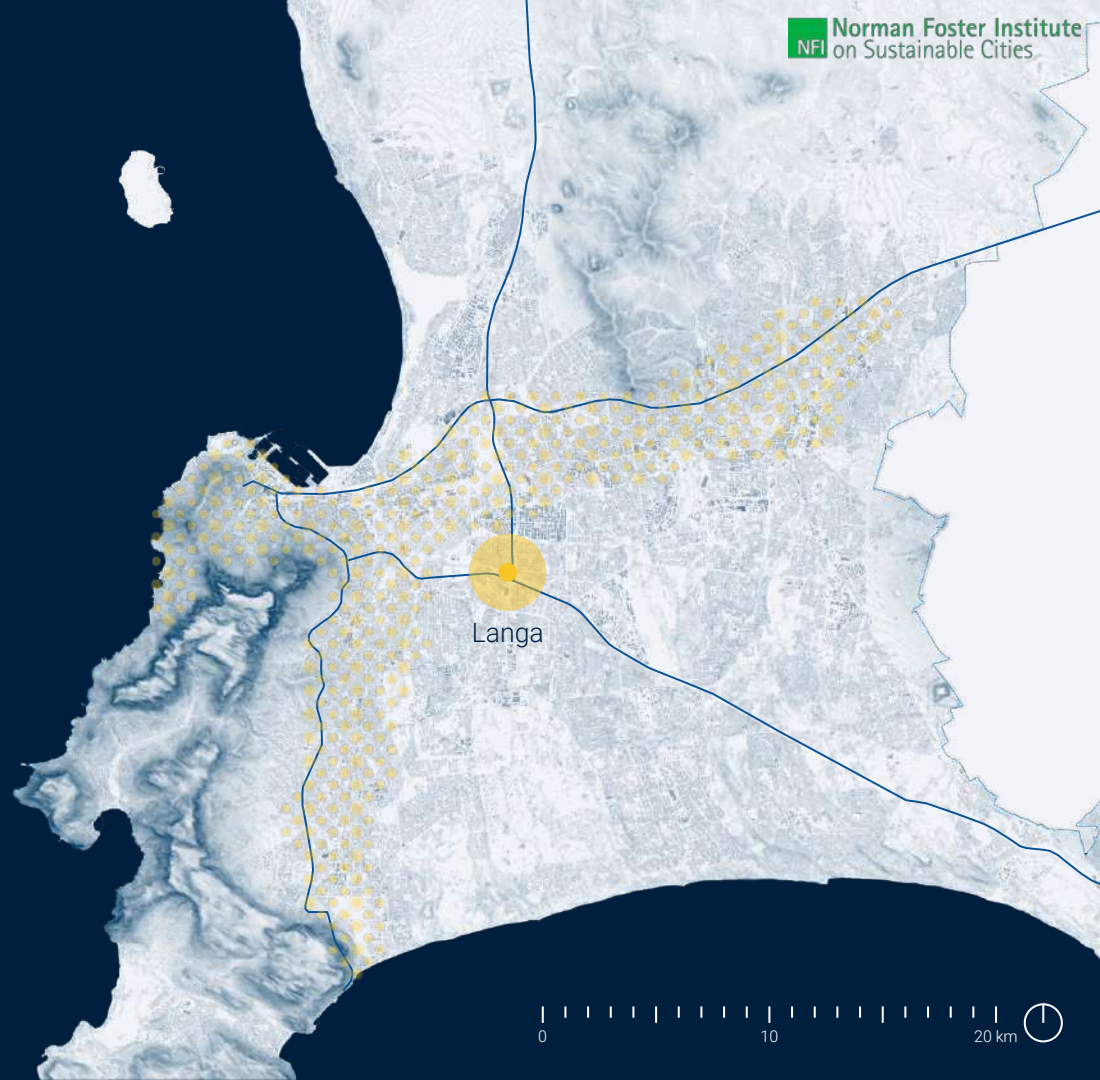
Additional Greenery

Space for Resting

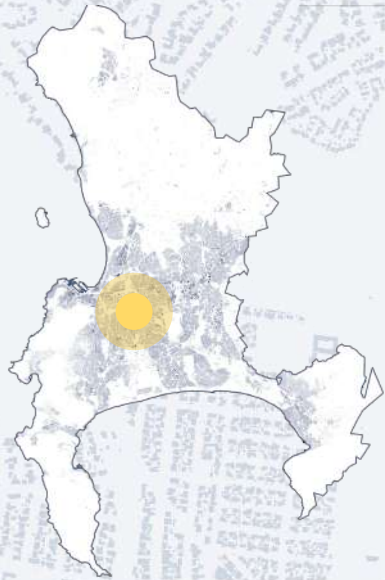
Public Square



Pilot Projects

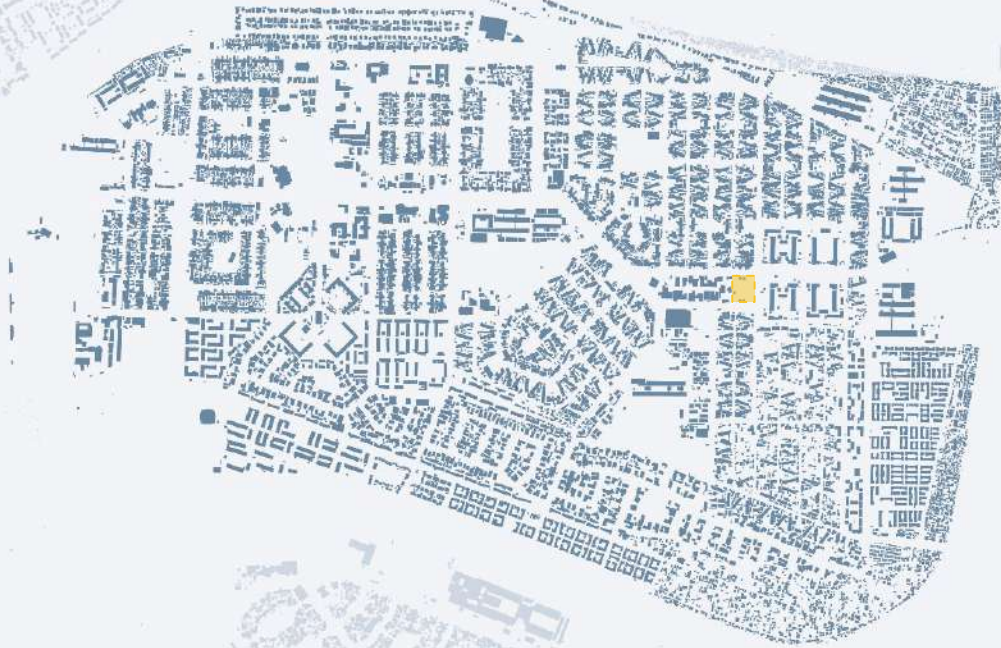


Pilot Project: Langa



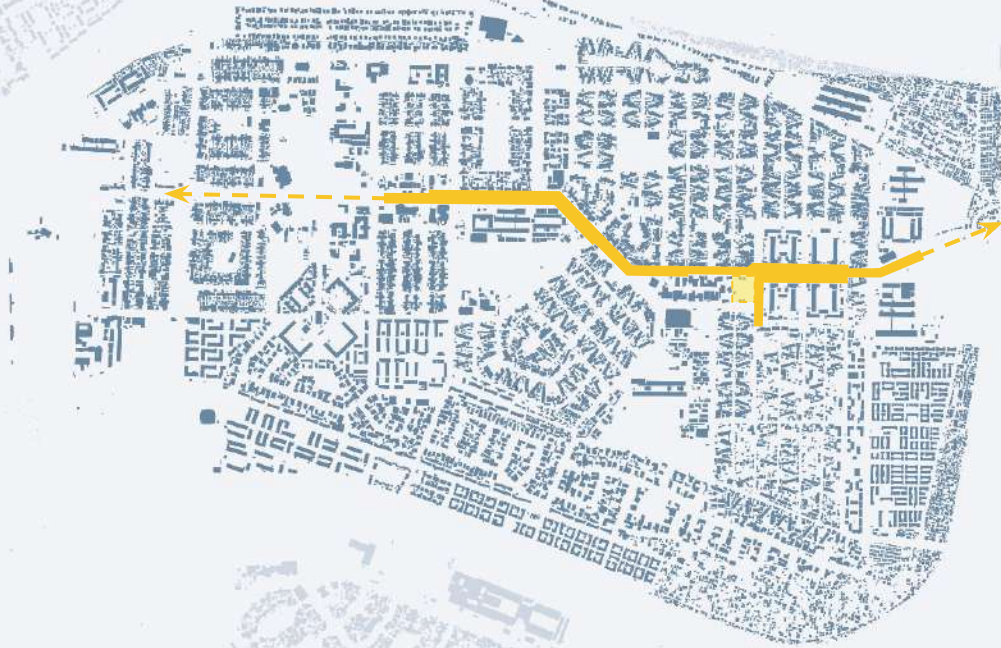
Pilot Project: Langa

Analysis for Site Selection: Transport Hubs (Metropolitan Scale Strategy)



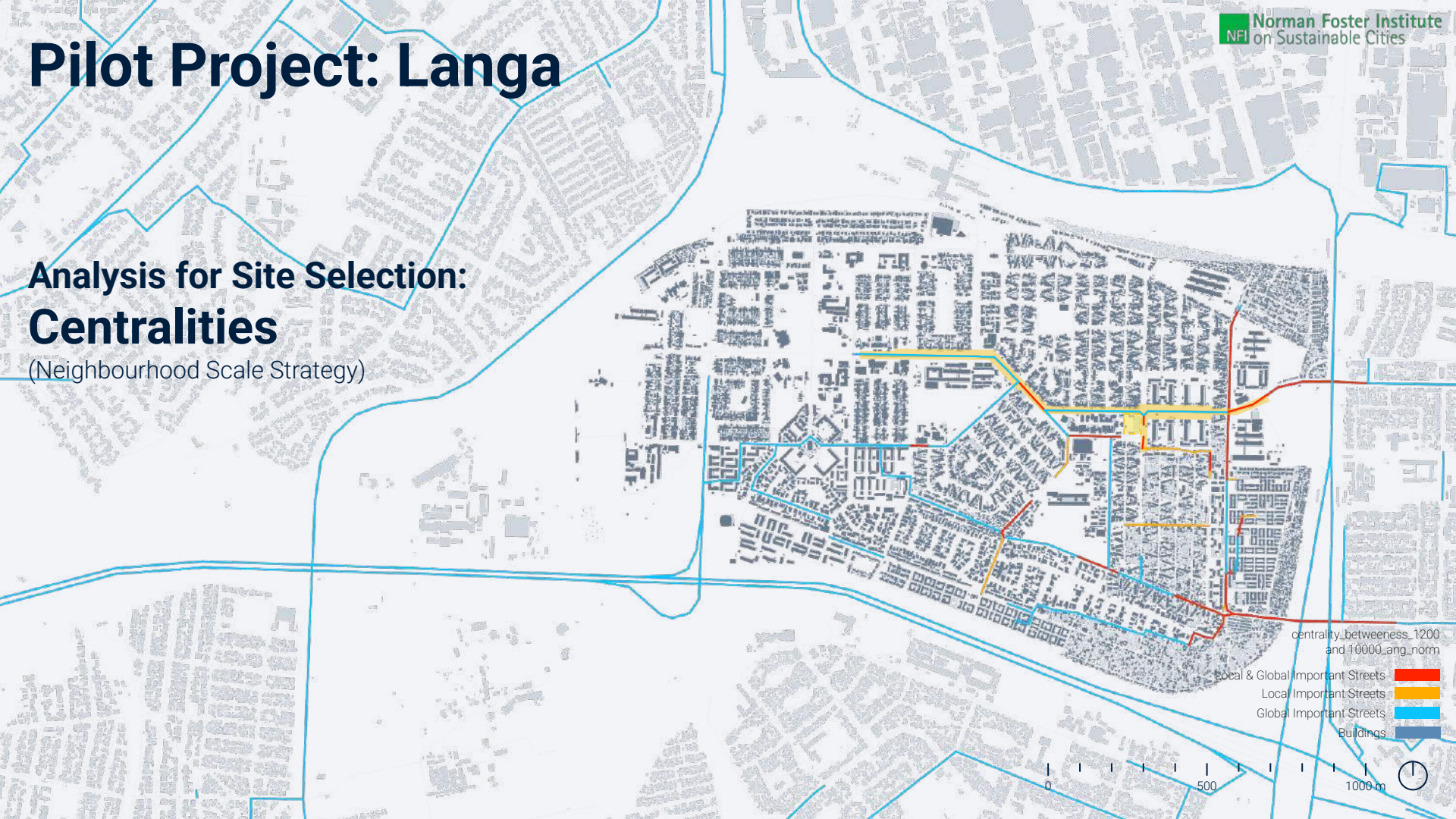
Pilot Project: Langa

Analysis for Site Selection: Thresholds (Metropolitan Scale Strategy)



Pilot Project: Langa

Analysis for Site Selection: Centralities (Neighbourhood Scale Strategy)



Pilot Project: Langa

Washington St

Washington Street



Pilot Project: Langa

Analysis for Programming & Design: Demographics

(Neighbourhood Scale Strategy)

70 000 Residents

99% Black African

25 Median Age

62% Youth Unemployment (15-24)



Pilot Project: Langa

Analysis for Programming & Design: Land Use Clusters

(Neighbourhood Scale Strategy)



Pilot Project: Langa

Analysis for Programming & Design: Safety

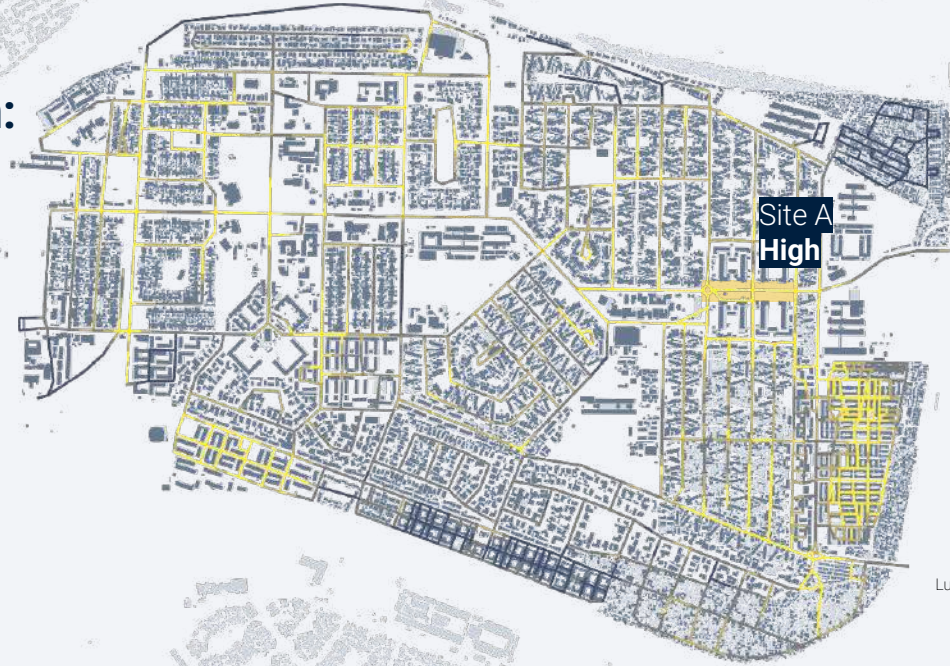
(Neighbourhood Scale Strategy)

Lumen Light Analysis

Visibility Analysis

Presence of Homeless

Porosity of Storefronts



Lumen Analysis Streets
decomposed 50m



Pilot Project: Langa

Analysis for Programming & Design: Walkability

(Neighbourhood Scale Strategy)

Access to Mini Bus Taxis

Access to Rail Network

Access to Bus

Points of Interest Analysis

Pedestrian/ Vehicle Conflict



Pilot Project: Langa

Analysis for Programming & Design: Comfort

(Neighbourhood Scale Strategy)

Access to Green Space

Percentage of Tree Coverage

Sky View Factor Analysis

Shadow Analysis

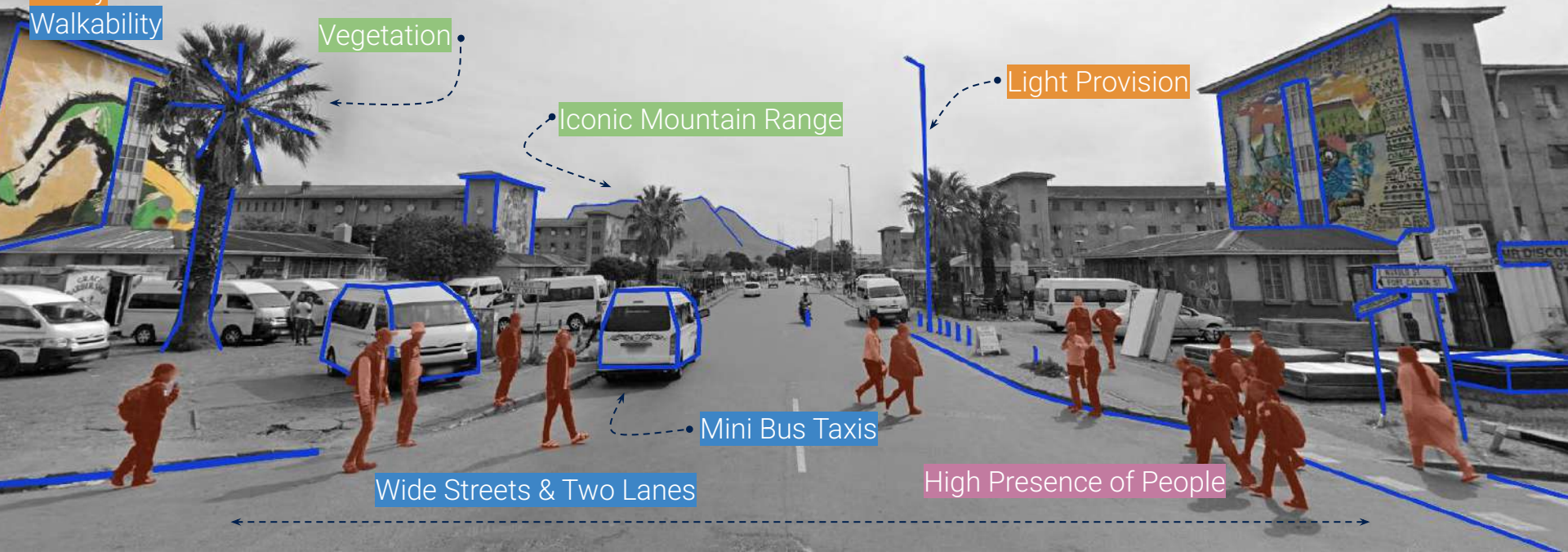
Presence of Urban Furniture



Pilot Project: Langa

Washington St

Demographics
Land Use Clusters
Comfort
Safety
Walkability



Vegetation

Iconic Mountain Range

Light Provision

Local Artist Mural

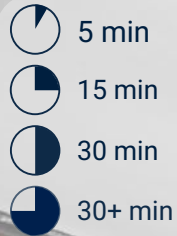
Mini Bus Taxis

Wide Streets & Two Lanes

High Presence of People

Pilot Project: Langa

Washington St



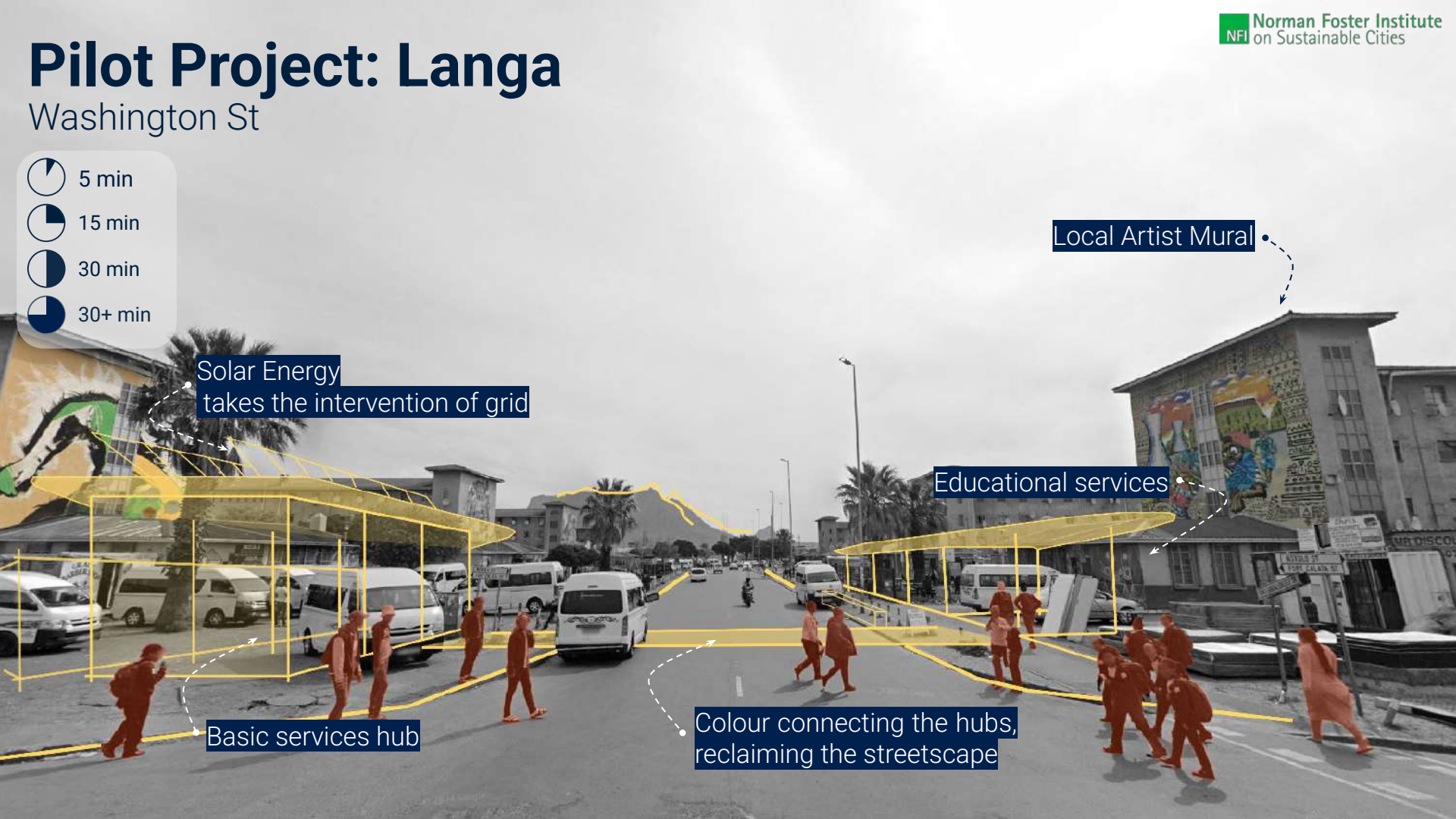
Solar Energy
takes the intervention of grid

Local Artist Mural

Educational services

Basic services hub

Colour connecting the hubs,
reclaiming the streetscape





4:30 pm



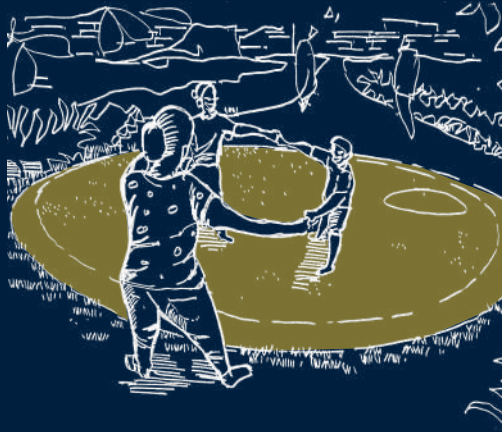
4:30 pm



5:45 pm



5:45 pm

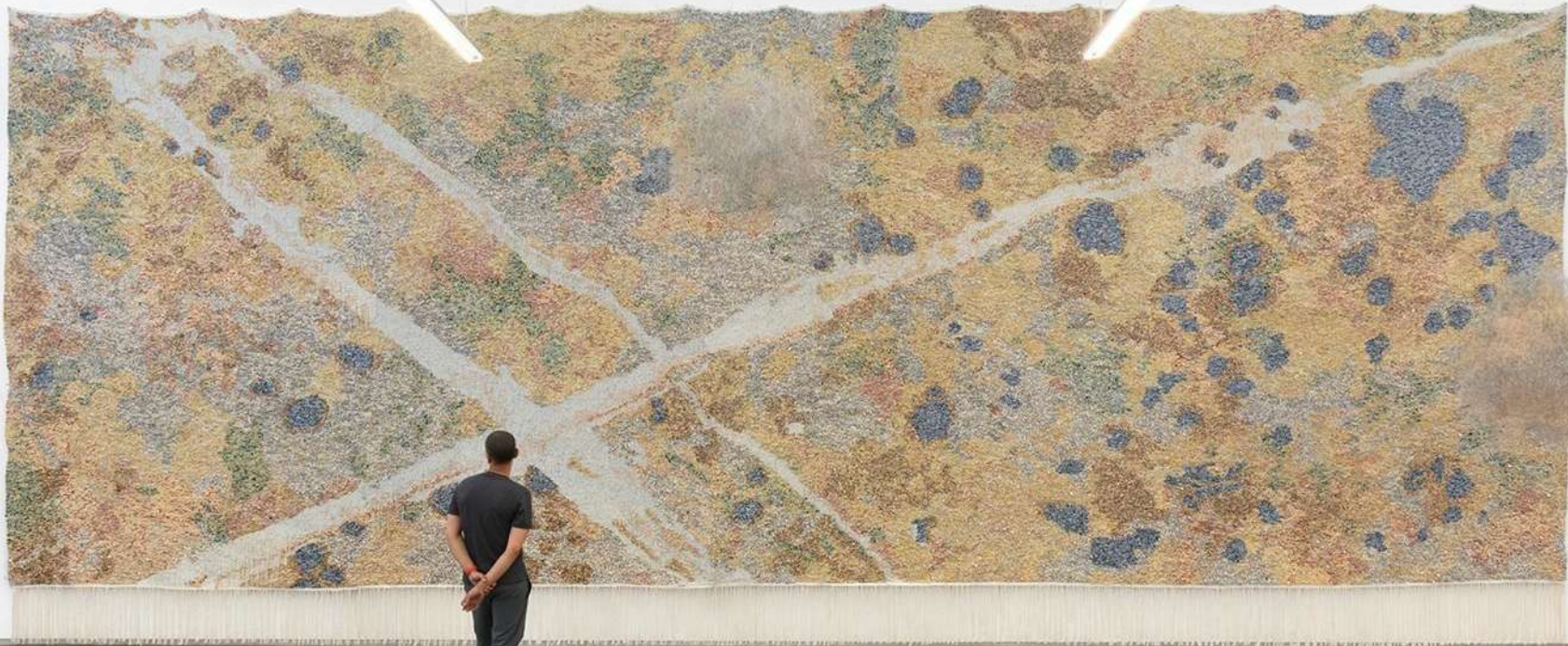


6:30 pm



6:30 pm

In waiting, **build**.
In Segregation, **knit**.



Bontehewuel, Igshaan Adams, 2021





Andrés

Architect
Spain



Dennis

Architect
Kenya



Mueni

Architect
Kenya



Nikhil

Data Scientist
Canada



Paulina

Architect
Germany



Enyine

Architectural Designer
Ghana



Josh

Entrepreneur
United States



Master On Sustainable Cities, 2025

Appendix 1: Ideas for a Business Model

How It's Operated



Community Runs It



NGO Supports It



City Scales It



How It's Funded

01 - Launch Phase

R500K
Public Private Partnership
Corporate Sponsors (adopt a hub)

02 - Community

Pilot Revenue
R5 Wi-Fi + Vendor fees

04 - Sustainable

Community Commerce
Local ads + scaled
vendor network

03 - City Adoption

After 18-month proof
Services Budget
Utilities + maintenance



Appendix 2: CBD and Langa Clustering Analysis

Content:

Pilot Project 1: CBD Commercial Cluster

Foreshore - Adderley - Strand

Commercial Cluster



Content:

Pilot Project 1: CBD Residential Clustering

Bo Kaap and District Six

Residential Clusters



Content:

Pilot Project 1: CBD Retail Cluster

Bree Street, Loop Street and Shortmarket Street

Retail Cluster



Content:

Pilot Project 2: Langa Educational Cluster

Central Langa and Washington Street

Education Cluster

